

where the perturbations are of the same order of magnitude as the unit roundoff u . This means that this method works as well as we could hope. The computed solution and residual satisfy (4.4.13) and (4.4.14) with $\epsilon_A \approx \epsilon_b \approx u$.

As Björck has shown (see, e.g., [7]), the modified Gram-Schmidt method is about as accurate as the QR decomposition by reflectors.

The method of normal equations has somewhat different characteristics. Suppose $A^T A$ and $A^T b$ have been calculated exactly, and the system $A^T A x = A^T b$ has been solved by Cholesky's method. From Section 2.7 (especially Exercise 2.7.29) we know that the computed solution is the exact solution of a perturbed equation $(A^T A + E)\hat{x} = A^T b$, where $\|E\|_2/\|A^T A\|_2$ is a modest multiple of u . It follows from Theorem 2.3.6 that $\|\delta x\|_2/\|x\|_2$ is roughly $\kappa_2(A^T A)u$. By Theorem 4.2.9, $\kappa_2(A^T A) = \kappa_2(A)^2$. Given that the factor $\kappa_2(A)^2$ also appears in (4.4.13), it looks as if the normal equations method is about as accurate as the other methods. However, in problems with a small residual, the $\kappa_2(A)^2 \tan \theta$ term in (4.4.13) is diminished. In these cases the other methods will be more accurate.

This analysis has assumed that $A^T A$ and $A^T b$ have been computed accurately. In practice significant errors can occur in the computation of $A^T A$, as was shown in Example 3.5.25. Furthermore, if $\cos \theta$ is near zero, cancellations will occur in the calculation of $A^T b$ and result in large relative errors.

Exercise 4.4.16 This exercise demonstrates that the QR method is superior to the normal equations method when the condition number is bad. It also shows how a change of basis can improve the condition number. Consider the following data.

t_i	y_i
0.98765431	2.1
0.98765432	2.1
0.98765433	2.1
0.98765434	2.0
0.98765435	1.9
0.98765436	1.9
0.98765437	1.8
0.98765438	1.7
0.98765439	1.7

- Set up an over-determined system $Ax = b$ for a linear polynomial. Use the basis $\phi_1(t) = 1$, $\phi_2(t) = t$. Using MATLAB, find the condition number of A . Find the least-squares solution by the QR decomposition method (e.g. $x = A \backslash b$ in MATLAB). Calculate the norm of the residual.
- Obtain the normal equations $A^T A x = A^T b$. Find the condition number of the coefficient matrix. Solve the normal equations. Calculate the norm of the residual ($\|b - Ax\|_2$).
- Compare your solutions from parts (a) and (b). Plot the two polynomials and the data points on a single graph.

- (d) Now set up a different over-determined system $\hat{A}\hat{x} = \hat{b}$ for the same data using the basis $\hat{\phi}_1(t) = 1$, $\hat{\phi}_2(t) = 3 \times 10^7 \times (t - .98765435)$. Find the condition number of $\hat{A}$. Find the least-squares solution by the QR decomposition method. Calculate the norm of the residual.
- (e) Obtain the normal equations $\hat{A}^T \hat{A} \hat{x} = \hat{A}^T \hat{b}$. Find the condition number of the coefficient matrix. Solve the normal equations. Calculate the norm of the residual ($\|\hat{b} - \hat{A}\hat{x}\|_2$).
- (f) Compare your solutions from parts (d) and (e). This time they are identical (to 15 decimal places if you care to check). What polynomial do these solutions represent? You might want to plot it just to make sure you've worked the problem correctly. If you do make a plot, you will see that your line looks identical to the one computed using the QR decomposition in part (a). That one was correct to only about eight decimal places, whereas the solutions computed using the well-conditioned matrices are correct to machine precision, but of course the difference does not show up in a plot (and wouldn't matter in most applications either).

□

Proof of Theorem 4.4.12 First of all, the perturbed problem has a unique solution, since $A + \delta A$ has full rank. This is a consequence of Corollary 4.2.16.

Rather than proving the theorem in full generality, we will examine the case in which only A is perturbed; that is, $\delta b = 0$. This will simplify the manipulations somewhat without altering the spirit of the proof. The complete proof is left as an exercise.

Furthermore, we shall make the assumption $\epsilon_A + \frac{1}{2}\epsilon_A^2 \leq 1/(4\kappa_2(A)^2)$, which is considerably more restrictive than the hypothesis that was given in the statement of the theorem. With a more careful argument we could remove this restriction.

The solution $\hat{x} = x + \delta x$ of the perturbed least squares problem must satisfy the perturbed normal equations

$$(A + \delta A)^T (A + \delta A)(x + \delta x) = (A + \delta A)^T b.$$

Subtracting the unperturbed normal equations $A^T A x = A^T b$ and making some routine manipulations, we find that

$$A^T A \delta x = \delta A^T (b - Ax) - A^T \delta A x - (A^T \delta A + \delta A^T A + \delta A^T \delta A) \delta x - \delta A^T \delta A x$$

or

$$\begin{aligned} \delta x &= (A^T A)^{-1} \delta A^T r - (A^T A)^{-1} A^T \delta A x \\ &\quad - (A^T A)^{-1} (A^T \delta A + \delta A^T A + \delta A^T \delta A) \delta x \\ &\quad - (A^T A)^{-1} \delta A^T \delta A x. \end{aligned} \tag{4.4.17}$$

Taking norms of both sides and using the fact that $\|B^T\|_2 = \|B\|_2$ for any B (Corollary 4.2.2), we obtain the inequality

$$\begin{aligned}\|\delta x\|_2 &\leq \|(A^T A)^{-1}\|_2 \|\delta A\|_2 \|r\|_2 + \|(A^T A)^{-1} A^T\|_2 \|\delta A\|_2 \|x\|_2 \\ &\quad + \|(A^T A)^{-1}\|_2 (2\|A\|_2 \|\delta A\|_2 + \|\delta A\|_2^2) \|\delta x\|_2 \\ &\quad + \|(A^T A)^{-1}\|_2 \|\delta A\|_2^2 \|x\|_2.\end{aligned}$$

As usual, let σ_1 and σ_m denote the largest and smallest singular values of A , respectively. Then $\|A\|_2 = \sigma_1$, $\|\delta A\|_2 = \sigma_1 \epsilon_A$, and from Theorem 4.2.11 we know that $\|(A^T A)^{-1}\|_2 = \sigma_m^{-2}$ and $\|(A^T A)^{-1} A^T\|_2 = \sigma_m^{-1}$. Furthermore, by (4.4.2) and (4.4.3), $\|r\|_2 = \|b\|_2 \sin \theta = \|Ax\|_2 \tan \theta \leq \sigma_1 \|x\|_2 \tan \theta$. Applying these results, making the substitution $\kappa_2(A) = \sigma_1/\sigma_m$, and moving all terms involving $\|\delta x\|_2$ to the left-hand side of the inequality, we obtain

$$\begin{aligned}[1 - 2\kappa_2(A)^2(\epsilon_A + \tfrac{1}{2}\epsilon_A^2)] \|\delta x\|_2 &\leq \\ &[\kappa_2(A)^2 \tan \theta + \kappa_2(A)] \epsilon_A \|x\|_2 + \kappa_2(A)^2 \epsilon_A^2 \|x\|_2.\end{aligned}$$

If $\epsilon_A + \frac{1}{2}\epsilon_A^2 \leq 1/(4\kappa_2(A)^2)$, then the coefficient of $\|\delta x\|_2$ is at least $1/2$. Therefore

$$\frac{\|\delta x\|_2}{\|x\|_2} \leq 2[\kappa_2(A)^2 \tan \theta + \kappa_2(A)] \epsilon_A + O(\epsilon_A^2).$$

This is (4.4.13) in the case $\epsilon_b = 0$.

To obtain the bound (4.4.14) on the residual, we note first that since $r + \delta r = b - (A + \delta A)(x + \delta x)$ and $r = b - Ax$,

$$\delta r = -\delta Ax - A\delta x - \delta A\delta x.$$

Using (4.4.17), and noting that by (4.4.13) $\|\delta A\|_2 \|\delta x\|_2 = O(\epsilon^2)$, we obtain

$$\begin{aligned}\|\delta r\|_2 &\leq \|\delta A\|_2 \|x\|_2 + \|A(A^T A)^{-1}\|_2 \|\delta A\|_2 \|r\|_2 \\ &\quad + \|A(A^T A)^{-1} A^T\|_2 \|\delta A\|_2 \|x\|_2 + O(\epsilon^2).\end{aligned}$$

Since $\|Ax\|_2 \geq \sigma_m \|x\|_2$, we have $\|x\|_2 \leq \sigma_m^{-1} \|Ax\|_2 = \sigma_m^{-1} \cos \theta \|b\|_2$. Also, by Theorem 4.2.11, $\|A(A^T A)^{-1}\|_2 = \sigma_m^{-1}$ and $\|A(A^T A)^{-1} A^T\|_2 = 1$. Therefore

$$\begin{aligned}\frac{\|\delta r\|_2}{\|b\|_2} &\leq (2 \cos \theta + \sin \theta) \kappa_2(A) \epsilon_A + O(\epsilon^2) \\ &< 3\kappa_2(A) \epsilon_A + O(\epsilon^2).\end{aligned}$$

This is just (4.4.14) in the case $\epsilon_b = 0$. □

Exercise 4.4.18 Work out a complete proof of Theorem 4.4.12 with $\epsilon_A > 0$ and $\epsilon_b > 0$. □

5

Eigenvalues and Eigenvectors I

Eigenvalues and eigenvectors turn up in stability theory, theory of vibrations, quantum mechanics, statistical analysis, and many other fields. It is therefore important to have efficient, reliable methods for computing these objects. The main business of this chapter is to develop such algorithms, culminating in the powerful and elegant *QR* algorithm.¹

Before we embark on the development of algorithms, we take the time to illustrate (in Section 5.1) how eigenvalues and eigenvectors arise in the analysis of systems of differential equations. The material is placed here entirely for motivational purposes. It is intended to convince you, the student, that eigenvalues are important. Section 5.1 is not, strictly speaking, a prerequisite for the rest of the chapter.

Section 5.1 also provides an opportunity to introduce MATLAB's `eig` command. When you use `eig` to compute the eigenvalues and eigenvectors of a matrix, you are using the *QR* algorithm.

5.1 SYSTEMS OF DIFFERENTIAL EQUATIONS

Many applications of eigenvalues and eigenvectors arise from the study of systems of differential equations.

¹The *QR algorithm* should not be confused with the *QR decomposition*, which we studied extensively in Chapter 3. As we shall see, the *QR algorithm* is an iterative procedure that performs *QR decompositions* repeatedly.

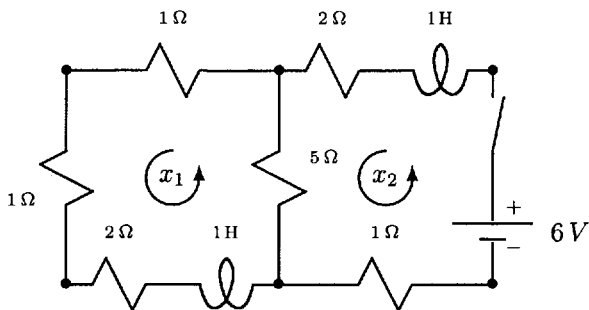


Fig. 5.1 Solve for the time-varying loop currents.

Example 5.1.1 The electrical circuit in Figure 5.1 is the same as the one that was featured in Example 1.2.8, except that two inductors and a switch have been added. Whereas resistors resist current, inductors resist changes in current. If we are studying constant, unvarying currents, as in Example 1.2.8, we can ignore the inductors, since their effect is felt only when the currents are changing. However, if the currents are varying in time, we must take the inductances into account.

Once the switch in the circuit is closed, current will begin to flow. The loop currents are functions of time: $x_i = x_i(t)$. Just as before, an equation for each loop can be obtained by applying Kirchhoff's voltage law: the sum of the voltage drops around the loop must be zero. The voltage drop across an inductor is proportional to the rate of change of the current. The constant of proportionality is the *inductance*. Thus if the current flowing through an inductor is $x(t)$ amps at time t seconds, and the inductance is m henries, the voltage drop across the inductor at time t is $m\dot{x}(t)$ volts, where $\dot{x}$ denotes the time derivative $\frac{dx}{dt}$ (amperes/second). Because of the presence of derivative terms, the resulting loop equations are now differential equations. Thus we have a system of two differential equations (one for each loop) in two unknowns (the loop currents).

Let us write down the two equations. First consider the first loop. As you will recall, the voltage drop across the 5Ω resistor in the direction indicated by the arrow for the first loop is $5(x_1 - x_2)$ volts. The voltage drop across the 1 henry inductor is $1\dot{x}_1$ volts. Summing these voltage drops, together with the voltage drops across the other resistors in loop 1, we obtain the equation

$$5(x_1 - x_2) + 1\dot{x}_1 + 1x_1 + 2x_1 + 1\dot{x}_1 = 0.$$

Similarly, in loop 2,

$$5(x_2 - x_1) + 1x_2 - 6 + 1\dot{x}_2 + 2x_2 = 0.$$

These are exactly the same as the equations we obtained in Example 1.2.8, except for the derivative terms. Rearranging these equations and employing matrix notation,

we can rewrite the system as

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -9 & 5 \\ 5 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 0 \\ -6 \end{bmatrix}. \quad (5.1.2)$$

Suppose the switch is closed at time zero. Then we can find the resulting currents by solving the system (5.1.2) subject to the initial conditions

$$\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (5.1.3)$$

As we shall see, we can solve this problem with the help of eigenvalues and eigenvectors. $\square$

The system of differential equations in Example 5.1.1 has the general form $\dot{x} = Ax - b$, where

$$x = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}, \quad A = \begin{bmatrix} -9 & 5 \\ 5 & -8 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 0 \\ -6 \end{bmatrix}.$$

A larger electrical circuit will lead to a larger system of equations of the same or similar form. For example, in Exercise 5.1.21 we consider a system of eight differential equations. The form is again $\dot{x} = Ax - b$, but now A is an 8×8 matrix. In general we will consider systems of n differential equations, in which the matrix A is $n \times n$.

Homogeneous Systems

An important step in solving a system of the form $\dot{x} = Ax - b$ is to solve the related *homogeneous* system $\dot{x} = Ax$, obtained by dropping the forcing term b (corresponding to the battery in Example 5.1.1). Thus we consider the problem of solving

$$\frac{dx}{dt} = Ax, \quad (5.1.4)$$

where A is an $n \times n$ matrix. Equation (5.1.4) is shorthand for

$$\begin{aligned} \frac{dx_1}{dt} &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ \frac{dx_2}{dt} &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ &\vdots \\ \frac{dx_n}{dt} &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n. \end{aligned}$$

A common approach to solving linear differential equations is to begin by looking for solutions of a particularly simple form. Therefore let us seek solutions of the form

$$x(t) = g(t)v, \quad (5.1.5)$$

where $g(t)$ is a nonzero scalar (real or complex valued) function of t , and v is a nonzero constant vector. The time-varying nature of $x(t)$ is expressed by $g(t)$, while the vector nature of $x(t)$ is expressed by v . Substituting form (5.1.5) into (5.1.4), we obtain the equation

$$\dot{g}(t)v = g(t)Av,$$

or equivalently

$$\frac{\dot{g}(t)}{g(t)}v = Av. \quad (5.1.6)$$

Since v and Av are constant vectors, (5.1.6) implies that $\dot{g}(t)/g(t)$ must be constant. That is, there exists a (real or complex) constant λ such that

$$\frac{\dot{g}(t)}{g(t)} = \lambda. \quad (5.1.7)$$

In addition (5.1.6) implies

$$Av = \lambda v.$$

A nonzero vector v for which there exists a λ such that $Av = \lambda v$ is called an *eigenvector* of A . The number λ is called the *eigenvalue* of A associated with v . So far we have shown that if $x(t)$ is a solution of (5.1.4) of the form (5.1.5), then v must be an eigenvector of A , and $g(t)$ must satisfy the differential equation (5.1.7), where λ is the eigenvalue of A associated with v . The general solution of the scalar differential equation (5.1.7) is $g(t) = ce^{\lambda t}$, where c is an arbitrary constant. Conversely, if v is an eigenvector of A with associated eigenvalue λ , then

$$e^{\lambda t}v$$

is a solution of (5.1.4), as you can easily verify. Thus each eigenvector of A gives rise to a solution of (5.1.4). If A has enough eigenvectors, then every solution of (5.1.4) can be realized as a linear combination of these simple solutions. Specifically, suppose A (which is $n \times n$) has a set of n linearly independent eigenvectors $v_1, \dots, v_n$ with associated eigenvalues $\lambda_1, \dots, \lambda_n$. Then for any constants $c_1, \dots, c_n$,

$$x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2 + \dots + c_n e^{\lambda_n t} v_n \quad (5.1.8)$$

is a solution of (5.1.4). Since $v_1, \dots, v_n$ are linearly independent, (5.1.8) turns out to be the general solution of (5.1.4); that is, every solution of (5.1.4) has the form (5.1.8). See Exercise 5.1.26.

An $n \times n$ matrix that possesses a set of n linearly independent eigenvectors is said to be a *semisimple* or *nondefective* matrix. In the next section we will see that in some sense “most” matrices are semisimple. Thus for “most” systems of the form (5.1.4), the general solution is (5.1.8).

It is easy to show (Section 5.2) that λ is an eigenvalue of A if and only if λ is a solution of the *characteristic equation* $\det(\lambda I - A) = 0$. For each eigenvalue λ a corresponding eigenvector can be found by solving the equation $(\lambda I - A)v = 0$. For

small enough (e.g. 2-by-2) systems of differential equations it is possible to solve the characteristic equation exactly and thereby solve the system.

Example 5.1.9 Find the general solution of

$$\dot{x}_1 = 2x_1 + 3x_2$$

$$\dot{x}_2 = x_1 + 4x_2.$$

The coefficient matrix is $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, whose characteristic equation is

$$\begin{aligned} 0 &= \det(\lambda I - A) = \det \begin{bmatrix} \lambda - 2 & -3 \\ -1 & \lambda - 4 \end{bmatrix} \\ &= (\lambda - 2)(\lambda - 4) - 3 = \lambda^2 - 6\lambda + 5 \\ &= (\lambda - 1)(\lambda - 5). \end{aligned}$$

Therefore the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 5$. Solving the equation $(\lambda I - A)v = 0$ twice, once with $\lambda = \lambda_1$ and once with $\lambda = \lambda_2$, we find (nonunique) solutions

$$v_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Since these two ($= n$) vectors are obviously linearly independent, we conclude that the general solution of the system is

$$x(t) = c_1 e^t \begin{bmatrix} 3 \\ -1 \end{bmatrix} + c_2 e^{5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (5.1.10)$$

□

Exercise 5.1.11 Check the details of Example 5.1.9.

□

The computations in Example 5.1.9 were quite easy because the matrix was only 2×2 , and the eigenvalues turned out to be integers. In a more realistic problem the system would be bigger, and the eigenvalues would be irrational numbers that we would have to approximate. Imagine trying to solve a system of the form $\dot{x} = Ax$ with, say, $n = 200$. We would need to find the 200 eigenvalues of a 200×200 matrix. Just trying to determine the characteristic equation (a polynomial equation of degree 200) is a daunting task, then we have to solve it. What is more, the problem of finding the zeros of a polynomial, given the coefficients, turns out to be ill conditioned whenever there are clustered roots, even when the underlying eigenvalue problem is well conditioned (Exercises 5.1.24 and 5.1.25). Clearly we need a better method. A major task of this chapter is to develop one.

Nonhomogeneous Systems

We now consider the problem of solving the nonhomogeneous system

$$\frac{dx}{dt} = Ax - b, \quad (5.1.12)$$

where b is a nonzero constant vector. Since b is invariant in time, it is reasonable to try to find a solution that is invariant in time: $x(t) = z$, where z is a constant vector. Substituting this form into (5.1.12), we have $dx/dt = 0$, so $0 = Az - b$, i.e. $Az = b$. Assuming A is nonsingular, we can solve the equation $Az = b$ to obtain a unique time-invariant solution z .

Once we have z in hand, we can make the following simple observation. If $y(t)$ is any solution of the homogeneous problem $\dot{x} = Ax$, then the sum $x(t) = y(t) + z$ is a solution of the nonhomogeneous system. Indeed $\dot{x} = \dot{y} + 0 = Ay = A(x - z) = Ax - Az = Ax - b$, so $\dot{x} = Ax - b$. Moreover, if $y(t)$ is the general solution of the homogeneous problem, then $x(t) = y(t) + z$ is the general solution of the nonhomogeneous problem.

Example 5.1.13 Let us solve the nonhomogeneous differential equation

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -9 & 5 \\ 5 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 0 \\ -6 \end{bmatrix}.$$

This is just (5.1.2) from Example 5.1.1. The two components of the solution, subject to the initial conditions (5.1.3), represent the two loop currents in the electrical circuit in Figure 5.1.

We must find a time-invariant solution and the general solution of the homogeneous problem, then add them together. First of all, the time-invariant solution must satisfy

$$\begin{bmatrix} -9 & 5 \\ 5 & -8 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -6 \end{bmatrix}.$$

This is exactly the system we had to solve in Example 1.2.8. The solution is $z_1 = 30/47 = 0.6383$ and $z_2 = 54/47 = 1.1489$. Recall that these numbers represent the loop currents in the circuit if inductances are ignored.

Now let us solve the homogeneous system

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -9 & 5 \\ 5 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

We can proceed as in Example 5.1.9 and form the characteristic equation $0 = \det(\lambda I - A) = \lambda^2 + 17\lambda + 47$, but this time the solutions are not integers. Applying the quadratic formula, we find that the solutions are $\lambda = (-17 \pm \sqrt{101})/2$. The rounded eigenvalues are $\lambda_1 = -13.5249$ and $\lambda_2 = -3.4751$. We can now substitute these values back into the equation $(\lambda I - A)v = 0$ to obtain eigenvectors. This is tedious by hand because the numbers are long decimals.

An easier approach is to let MATLAB do the work. Using MATLAB's `eig` command, we obtain the (rounded) eigenvectors

$$v_1 = \begin{bmatrix} 0.7415 \\ -0.6710 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 0.6710 \\ 0.7415 \end{bmatrix}.$$

Thus the general solution to the homogeneous problems is

$$c_1 e^{-13.5249t} \begin{bmatrix} 0.7415 \\ -0.6710 \end{bmatrix} + c_2 e^{-3.4751t} \begin{bmatrix} 0.6710 \\ 0.7415 \end{bmatrix}.$$

Adding this to the time-invariant solution of the nonhomogeneous problem, we obtain the general solution of the nonhomogeneous problem:

$$x(t) = \begin{bmatrix} 0.6383 \\ 1.1489 \end{bmatrix} + c_1 e^{-13.5249t} \begin{bmatrix} 0.7415 \\ -0.6710 \end{bmatrix} + c_2 e^{-3.4751t} \begin{bmatrix} 0.6710 \\ 0.7415 \end{bmatrix}. \quad (5.1.14)$$

To obtain the particular solution that gives the loop currents in the circuit, assuming the switch is closed at time zero, we must apply the initial condition (5.1.3), $x(0) = 0$. Making this substitution in (5.1.14), we obtain

$$0 = \begin{bmatrix} 0.6383 \\ 1.1489 \end{bmatrix} + c_1 \begin{bmatrix} 0.7415 \\ -0.6710 \end{bmatrix} + c_2 \begin{bmatrix} 0.6710 \\ 0.7415 \end{bmatrix},$$

which can be seen to be a system of two linear equations that we can solve for c_1 and c_2 . Doing so (using MATLAB) we obtain

$$c_1 = 0.2977 \quad \text{and} \quad c_2 = -1.2802.$$

Using this solution we can determine what the loop currents will be at any time.

We bring this example to a close by considering the long-term behavior of the circuit. Examining (5.1.14), we see that the exponential functions $e^{-13.5249t}$ and $e^{-3.4751t}$ both tend to zero as $t \rightarrow \infty$, since the exponents are negative. This is a consequence of both eigenvalues $\lambda_1 = -13.5249$ and $\lambda_2 = -3.4751$ being negative. In fact the convergence to zero is quite rapid in this case. After a short time, the exponential functions become negligible and the solution is essentially constant: $x(t) \approx \begin{bmatrix} 0.6383 \\ 1.1489 \end{bmatrix}$. In other words, after a brief transient phase, the circuit settles down to its steady state. Figure 5.2 shows a plot of the loop currents as a function of time during the first second after the switch has been closed. Notice that after just one second the currents are already quite close to their steady-state values. $\square$

A larger circuit is studied in Exercise 5.1.21. More interesting circuits can be built by inserting some capacitors and thereby obtaining solutions that oscillate in time. Of course, mass-spring systems also exhibit oscillatory behavior.

Dynamics of Simple Mass-Spring Systems

Figure 5.3 depicts a cart attached to a wall by a spring. At time zero the cart is at rest at its equilibrium position $x = 0$. At that moment a steady force of 2 newtons

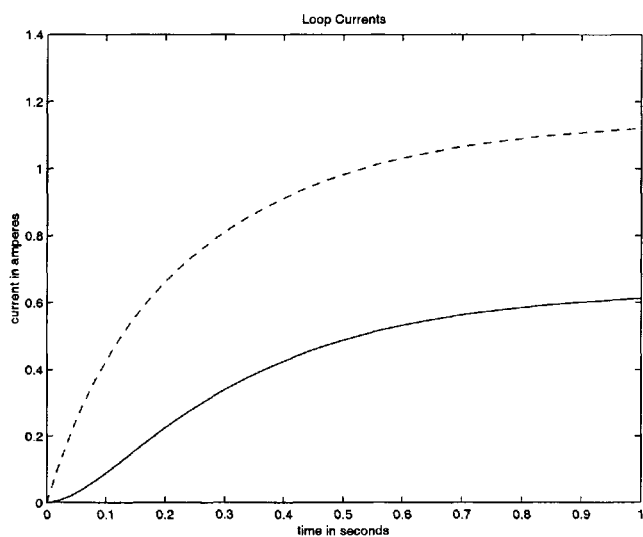


Fig. 5.2 Loop currents as a function of time

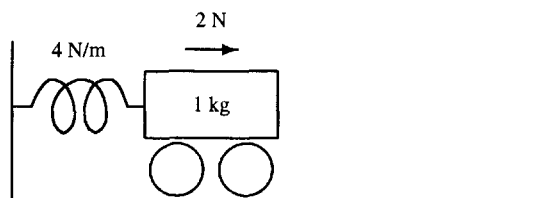


Fig. 5.3 Single cart and spring

is applied, pushing the cart to the right and stretching the spring. In Section 1.2 we studied this same system (Figure 1.3) and found that the new equilibrium position is $x = 0.5$ meter. That is, the force moves the cart one half meter to the right. Now we will ask the more specific question of how the cart gets from its old equilibrium position to the new one. The solution of the static problem does not depend on the mass of the cart, but the dynamic problem does. The rolling friction also matters. Let us say that the mass of the cart is 1 kilogram. We'll make the linear assumption that the rolling friction is proportional to the velocity: $-k \dot{x}(t)$ newtons, where k is a proportionality constant with units kg/s.

The physical principle that we will apply is Newton's second law, $f = ma$, in the form $a = f/m$: the acceleration of the cart is equal to the sum of the forces on the cart divided by the cart's mass. The choice $m = 1$ simplifies things. The three forces on the cart are the external force $+2$ N, the restoring force of the spring $-4x(t)$ N, and the rolling friction $-k \dot{x}(t)$ N. We recall also that the acceleration is $\ddot{x}$, the second derivative of displacement with respect to time. Thus Newton's second law gives the differential equation

$$\ddot{x} = 2 - 4x - k\dot{x}, \quad (5.1.15)$$

which we can solve subject to the initial conditions $x(0) = \dot{x}(0) = 0$ to find x . It turns out that the nature of the motion depends heavily on the value of k .

The differential equation (5.1.15) is simple enough that we can solve it as it stands using standard techniques from a first course in differential equations. However, we will follow a different course. The differential equation is of second order; that is, it involves a second derivative. We will convert it to a system of two first-order differential equations by a standard technique. Let $x_1(t) = x(t)$ and $x_2(t) = \dot{x}(t)$. Then we can rewrite the differential equation as $\dot{x}_2 = 2 - 4x_1 - kx_2$. Combining this with the trivial differential equation $\dot{x}_1 = x_2$, we obtain the following system and initial condition:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -k \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \quad \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This system has the general form $\dot{x} = Ax - b$, just as our system in the electrical circuit example did, and it can be solved by the same means. A system of this type is solved in Exercise 5.1.22.

A system of three masses attached by springs is considered in Exercise 5.1.23. In that problem Newton's second law is applied to each of the carts to obtain a system of three second-order differential equations, which is then rewritten as a system of six first-order differential equations.

Stability of Linear and Nonlinear Systems

In stability theory the behavior of systems as $t \rightarrow \infty$ is studied. In Example 5.1.13 we noted that as $t \rightarrow \infty$ the loop currents tend to their equilibrium or steady-state values. This is so because the exponents in the general solution (5.1.14) are negative, which is in turn a consequence of the eigenvalues being negative. It holds regardless

of the values of the constants c_1 and c_2 , which means that it holds for all solutions. Thus if the loop currents are somehow perturbed away from their equilibrium values, they will return to those values. This is *stable* behavior.

The system in Example 5.1.9, on the other hand, behaves quite differently. There the eigenvalues are positive, so the exponents in the general solution (5.1.10) are positive and become large as $t \rightarrow \infty$. Consequently all solutions become large as time increases. The only exception is the steady-state solution 0, obtained by setting $c_1 = c_2 = 0$. If the system is initially in this equilibrium state but then is somehow perturbed away from it, it will not return to that state. It will blow up, because all nonzero solutions blow up. This is *unstable* behavior.

In general we can assess the stability of homogeneous linear systems (5.1.4) by examining the general solution (5.1.8). (In the nonhomogeneous case we just add a constant.) If all of the eigenvalues λ_k are real and negative, then all solutions will satisfy $x(t) \rightarrow 0$ as $t \rightarrow \infty$. On the other hand, if at least one of the eigenvalues is positive, then there will be solutions that satisfy $\|x(t)\| \rightarrow \infty$ as $t \rightarrow \infty$. In the former case the system is said to be *asymptotically stable*; in the latter case it is *unstable*.

As we shall see, eigenvalues need not always be real numbers. Even if A is real, some or all of its eigenvalues can be complex. What happens then? Say $\lambda_k = \alpha_k + i\beta_k$. Then $e^{\lambda_k t} = e^{\alpha_k t} e^{i\beta_k t} = e^{\alpha_k t} (\cos \beta_k t + i \sin \beta_k t)$, which grows if $\alpha_k > 0$ and decays if $\alpha_k < 0$. Since $\alpha_k = \operatorname{Re}(\lambda_k)$ (the *real part* of λ_k), we conclude that (5.1.4) is asymptotically stable if all eigenvalues of A satisfy $\operatorname{Re}(\lambda_k) < 0$, and it is unstable if one or more of the eigenvalues satisfies $\operatorname{Re}(\lambda_k) > 0$. We have come to this conclusion under the assumption that A is a semisimple matrix. However, the conclusion turns out to be valid for all A .

Notice that complex eigenvalues signal oscillations in the solution, since they give rise to sine and cosine terms. The frequency of oscillation of the k th term is governed by $\beta_k = \operatorname{Im}(\lambda_k)$. See Exercises 5.1.22 and 5.1.23 for examples of systems that have complex eigenvalues and therefore exhibit oscillatory behavior.

Now consider a nonlinear system of n first-order differential equations

$$\begin{aligned}\dot{x}_1 &= f_1(x_1, x_2, \dots, x_n) \\ \dot{x}_2 &= f_2(x_1, x_2, \dots, x_n) \\ &\vdots \\ \dot{x}_n &= f_n(x_1, x_2, \dots, x_n)\end{aligned}$$

or, in vector notation,

$$\dot{x} = f(x). \quad (5.1.16)$$

A vector $\hat{x}$ is called an *equilibrium point* of (5.1.16) if $f(\hat{x}) = 0$, since then the constant function $x(t) = \hat{x}$ is a solution. An equilibrium point $\hat{x}$ is called *asymptotically stable* if there is a neighborhood N of $\hat{x}$, such that for every solution of (5.1.16) for which $x(t) \in N$ for some t , it must happen that $x(t) \rightarrow \hat{x}$ as $t \rightarrow \infty$. Physically this means that a system whose state is perturbed slightly from an equilibrium point will return to that equilibrium point. We can analyze the stability of each equilibrium point by linearizing (5.1.16) about that point.

Specifically, consider a multivariate Taylor expansion of f centered on $\hat{x}$:

$$\begin{aligned} f_i(x_1, \dots, x_n) &= f_i(\hat{x}) + \sum_{j=1}^n \frac{\partial f_i}{\partial x_j}(\hat{x})(x_j - \hat{x}_j) \\ &\quad + \frac{1}{2} \sum_{j=1}^n \sum_{k=1}^n \frac{\partial^2 f_i}{\partial x_j \partial x_k}(\hat{x})(x_j - \hat{x}_j)(x_k - \hat{x}_k) + \dots \end{aligned}$$

for $i = 1, \dots, n$. In vector notation we have the more compact expression

$$f(x) = f(\hat{x}) + \frac{\partial f}{\partial x}(\hat{x})(x - \hat{x}) + \text{higher order terms},$$

where $\partial f / \partial x$ is the *Jacobian matrix*, whose (i, j) entry is $\partial f_i / \partial x_j$. For x near $\hat{x}$, the quadratic and higher terms are very small. Also $f(\hat{x}) = 0$, so $f(x) \approx \frac{\partial f}{\partial x}(\hat{x})(x - \hat{x})$. It is thus reasonable to expect that for x near $\hat{x}$ the solutions of (5.1.16) are approximated well by solutions of

$$\frac{d}{dt}(x - \hat{x}) = \dot{x} = \frac{\partial f}{\partial x}(\hat{x})(x - \hat{x}). \quad (5.1.17)$$

This is a linear system with coefficient matrix $\frac{\partial f}{\partial x}(\hat{x})$. From our discussion of stability of linear systems we know that all solutions of (5.1.17) satisfy $x - \hat{x} \rightarrow 0$ as $t \rightarrow \infty$ if all eigenvalues of $\frac{\partial f}{\partial x}(\hat{x})$ satisfy $\operatorname{Re}(\lambda_k) < 0$. We conclude that $\hat{x}$ is an asymptotically stable equilibrium point of (5.1.16) if all eigenvalues of the Jacobian $\frac{\partial f}{\partial x}(\hat{x})$ have negative real parts.

Exercise 5.1.18 The differential equation of a damped pendulum is

$$\ddot{\theta} + k_1 \dot{\theta} + k_2 \sin \theta = 0,$$

where k_1 and k_2 are positive constants, and θ is the angle of the pendulum from its vertical resting position. This differential equation is nonlinear because $\sin \theta$ is a nonlinear function of θ . Introducing new variables $x_1 = \theta$ and $x_2 = \dot{\theta}$, rewrite the second-order differential equation as a system of two first-order differential equations in x_1 and x_2 . Find all equilibrium points of the system, and note that they correspond to $\dot{\theta} = 0$, $\theta = n\pi$, $n = 0, \pm 1, \pm 2, \dots$. Calculate the Jacobian of the system and show that the equilibrium points $\theta = n\pi$ are asymptotically stable for even n and unstable for odd n . Interpret your result physically. $\square$

Additional Exercises

Exercise 5.1.19 Suppose A is nonsingular and has linearly independent eigenvectors $v_1, \dots, v_n$. Let V be the nonsingular $n \times n$ matrix whose columns are $v_1, \dots, v_n$.

- (a) Show that (5.1.8) can be rewritten as $x(t) = V e^{\Lambda t} c$, where c is a column vector, Λt is the diagonal matrix $\operatorname{diag}\{\lambda_1 t, \dots, \lambda_n t\}$, and $e^{\Lambda t}$ is its *matrix exponential*: $e^{\Lambda t} = \operatorname{diag}\{e^{\lambda_1 t}, \dots, e^{\lambda_n t}\}$.

- (b) Show that the general solution of $\dot{x} = Ax - b$ has the form $x(t) = z + Ve^{\Lambda t}c$, where z satisfies $Az = b$.
- (c) To solve the initial value problem $\dot{x} = Ax - b$ with initial condition $x(0) = \hat{x}$, we need to solve for the constants in the vector c . Show that c can be obtained by solving the system $Vc = \hat{x} - z$. Since V is nonsingular, this system has a unique solution.

□

Exercise 5.1.20 Using MATLAB, work out the details of Example 5.1.13. The MATLAB command `[V,D]=eig(A)` returns (if possible) a matrix V whose columns are linearly independent eigenvectors of A and a diagonal matrix D whose main diagonal entries are the eigenvalues of A . Thus V and D are the same as the matrices V and Λ of the previous exercise. You may find the results of the previous exercise helpful as you work through this exercise. Here are some sample plot commands:

```
t = 0:.02:1;
x = z*ones(size(t));
for j=1:2; x = x + V(:,j)*c(j)*exp(t*D(j,j)); end
plot(t,x(1,:),t,x(2,:), '--')
title('Loop Currents')
xlabel('time in seconds')
ylabel('current in amperes')
print
```

Remember that for more information on the usage of these commands, you can type `help plot`, `help print`, etc. □

Exercise 5.1.21 In working through this exercise, do not overlook the advice given in the previous two exercises. The circuit in Figure 5.4 is the same as the one that appeared in Exercise 1.2.19, except that some inductors have been added around the edges. All of the resistances are $1\ \Omega$, and all of the inductances are $1\ \text{H}$. Notice the unusual numbering of the loops. Initially the loop currents are all zero, because the switch is open. Suppose the switch is closed at time 0.

- Write down a system of nine equations for the nine unknown loop currents. All except the ninth equation involve derivatives.
- Solve the ninth equation for x_9 in terms of other loop currents, and use this expression to eliminate x_9 from the other equations. Obtain thereby a system of eight differential equations $\dot{x} = Ax - b$ with initial condition $x(0) = 0$.
- Solve the system $Az = b$ to obtain a steady-state solution of the differential equation. Compute the steady-state value of the loop current z_9 also. (Checkpoint: $z_1 = 0.3$ amperes.)
- Get the eigenvalues and eigenvectors of A , and use them to construct the general solution of the homogeneous equation $\dot{z} = Az$.
- Deduce the solution of the initial value problem $\dot{x} = Ax - b$, $x(0) = 0$.

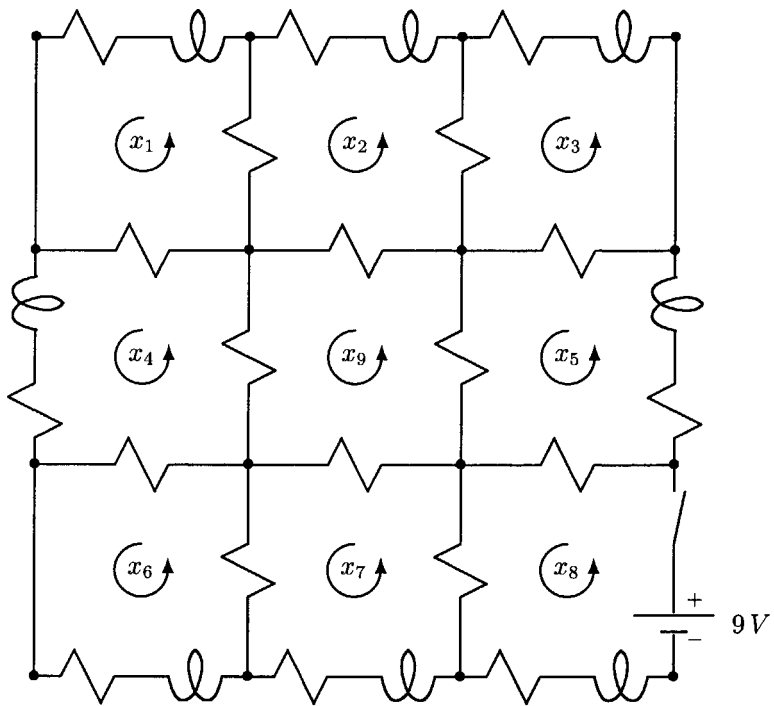


Fig. 5.4 Solve for the time-varying loop currents.

- (f) On separate graphs plot the loop currents $x_1(t)$ and $x_8(t)$ for $0 \leq t \leq 6$ seconds. (Checkpoint: Do these look realistic?)
- (g) Once the switch is closed, about how long does it take for all of the loop currents to be within (i) ten percent of their final values? (ii) one percent?

□

Exercise 5.1.22 Consider a cart attached to a wall by a spring, as shown in Figure 5.5. At

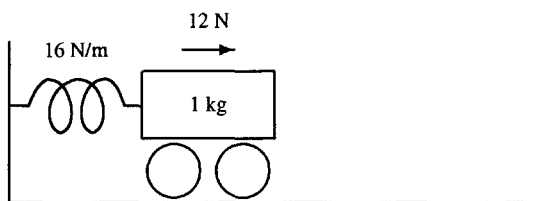


Fig. 5.5 Solve for the motion of the cart.

time zero the cart is at rest at its equilibrium position $x = 0$. At that moment a steady force of 12 newtons is applied, pushing the cart to the right. Assume that the rolling friction is $-k \dot{x}(t)$ newtons. Do parts (a) through (d) by hand.

- Set up a system of two first-order differential equations of the form $\dot{x} = Ax - b$ for $x_1(t) = x(t)$ and $x_2(t) = \dot{x}(t)$.
- Find the steady-state solution of the differential equation.
- Find the characteristic equation of A and solve it by the quadratic formula to obtain an expression (involving k) for the eigenvalues of A .
- There is a *critical value* of k at which the eigenvalues of A change from real to complex. Find this critical value.
- Using MATLAB, solve the initial value problem for the cases (i) $k = 2$, (ii) $k = 6$, (iii) $k = 10$, and (iv) $k = 14$. Rather than reporting your solutions, simply plot $x_1(t)$ for $0 \leq t \leq 3$ for each of your four solutions on a single set of axes. (Do not overlook the help given in Exercises 5.1.19 and 5.1.20.) Comment on your plots (e.g. rate of decay to steady state, presence or absence of oscillations).
- What happens when $k = 0$?

□

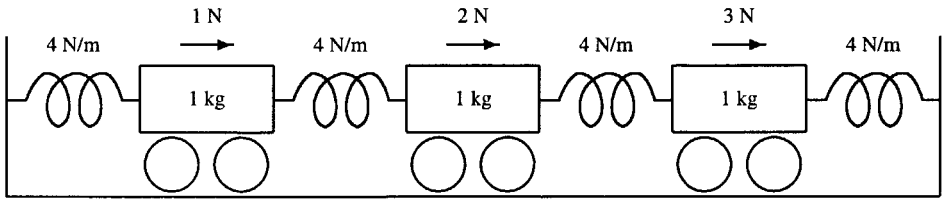


Fig. 5.6 Solve for the motion of the carts.

Exercise 5.1.23 Consider a system of three carts attached by springs, as shown in Figure 5.6. The carts are initially at rest. At time zero the indicated forces are applied, causing the carts to move toward a new equilibrium. Let x_1 , x_2 , and x_3 denote the displacements of the carts, and let x_4 , x_5 , and x_6 denote their respective velocities. Suppose the coefficients of rolling friction of the three carts are k_1 , k_2 , and k_3 , respectively.

- Apply Newton's second law to each cart to obtain a system of three second-order differential equations for the displacements x_1 , x_2 , and x_3 . You may find it useful to review Example 1.2.10.
- Introducing the velocity variables x_4 , x_5 , and x_6 , rewrite your system as a system of six first-order differential equations. Write your system in the form $\dot{x} = Ax - b$.
- Find the steady-state solution of the system.
- Solve the initial value problem under each of the conditions listed below. In each case plot x_1 , x_2 , and x_3 on a single set of axes for $0 \leq t \leq 20$, and comment on the plot.
 - $k_1 = 1$, $k_2 = 0$, and $k_3 = 0$.
 - $k_1 = 1$, $k_2 = 8$, and $k_3 = 8$.
 - $k_1 = 8$, $k_2 = 8$, and $k_3 = 8$.

□

Exercise 5.1.24 Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, the 2×2 identity matrix.

- Show that the characteristic equation of A is

$$\lambda^2 - 2\lambda + 1 = 0.$$

The eigenvalues are obviously $\lambda_1 = 1$ and $\lambda_2 = 1$.

- (b) We now perturb one coefficient of the characteristic polynomial slightly and consider the equation

$$\lambda^2 - 2\lambda + (1 - \epsilon) = 0,$$

where $\epsilon \ll 1$. Solve the equation for the roots $\hat{\lambda}_1$ and $\hat{\lambda}_2$.

- (c) Show that when $\epsilon = 10^{-12}$, $|\hat{\lambda}_1 - \lambda_1|$ and $|\hat{\lambda}_2 - \lambda_2|$ are one million times bigger than ϵ .
- (d) Sketch the graphs of the original and perturbed polynomials (using some ϵ bigger than 10^{-12} , obviously), and give some indication of why the roots are so sensitive to the ϵ perturbation.

This shows that the problem of finding the roots of a polynomial equation can be ill conditioned. Difficulties arise any time there are multiple roots or tightly clustered roots. The more roots there are in the cluster, the worse the situation is. In contrast, the *eigenvalue problem* for the identity matrix is well conditioned. Small perturbations in the entries of A cause small changes in the eigenvalues. This is true not only for the identity matrix but for any symmetric matrix, regardless of whether or not the eigenvalues are clustered, as we shall show in Section 6.5 (Corollary 6.5.8). $\square$

Exercise 5.1.25 Figure out what the following MATLAB commands do. For example, type `help poly` in MATLAB to find out what `poly` does.

```
A = eye(2)
lam = eig(A)
c = poly(A)
roots(c)
format long
A2 = A + 200*eps*randn(2)
lam2 = eig(A2)
eigdif = norm(lam2-lam)
c2 = c + [0 200*eps*randn(1,2)]
roots2 = roots(c2)
rootdif = norm(roots2-lam)
```

Run this code several times. The results will vary because random numbers are being used. Explain what the output shows. $\square$

Exercise 5.1.26 In this exercise we show that (5.1.8) is the general solution of (5.1.4).

- (a) Show that for any $\hat{x} \in \mathbb{C}^n$ there exist (unique) $c_1, \dots, c_n$ such that the function $x(t)$ of (5.1.8) satisfies $x(0) = \hat{x}$.
- (b) It is a basic fact of the theory of differential equations that for a given $\hat{x} \in \mathbb{C}^n$, the *initial value problem* $dx/dt = Ax$, $x(0) = \hat{x}$, has exactly one solution (see any text on differential equations). Use this fact together with the result of part (a) to show that every solution of (5.1.4) has the form (5.1.8).

$\square$

5.2 BASIC FACTS ABOUT EIGENVALUES AND EIGENVECTORS

In most scientific and engineering applications that require eigenvalue computations, the matrix has real entries. Nevertheless, as we shall soon see, the natural setting for the study of eigenvalues is the field of complex numbers. Thus let $A \in \mathbb{C}^{n \times n}$. A vector $v \in \mathbb{C}^n$ is called an *eigenvector* of A if $v \neq 0$ and Av is a multiple of v ; that is, there exists a $\lambda \in \mathbb{C}$ such that

$$Av = \lambda v. \quad (5.2.1)$$

The scalar λ is called the *eigenvalue* of A associated with the eigenvector v . Similarly v is called an eigenvector of A associated with the eigenvalue λ . The pair (λ, v) is called an *eigenpair* of A . Whereas each eigenvector has a unique eigenvalue associated with it, each eigenvalue is associated with many eigenvectors. For example, if v is an eigenvector of A associated with the eigenvalue λ , then every nonzero multiple of v is also an eigenvector of A associated with λ .

The set of all eigenvalues of A is called the *spectrum* of A .

Frequently we will wish to work with eigenvectors with unit norm. This is always possible, for given any eigenvector v of A , the vector $\hat{v} = (1/\|v\|)v$ is also an eigenvector of A associated with the same eigenvalue, and $\|\hat{v}\| = 1$. This normalization process can be carried out using any vector norm.

Exercise 5.2.2 Show that A is nonsingular if and only if 0 is not an eigenvalue of A . (Thus we could add another line to the list in Theorem 1.2.3.) $\square$

In the previous section we mentioned that λ is an eigenvalue of A if and only if it is a root of the characteristic equation. Let us now see why this is true. The equation (5.2.1) can be rewritten as $Av = \lambda Iv$ or $(\lambda I - A)v = 0$, where I is the $n \times n$ identity matrix. Thus if v is an eigenvector with eigenvalue λ , then v is a nonzero solution of the homogeneous matrix equation $(\lambda I - A)v = 0$. Therefore, by part (b) of Theorem 1.2.3, the matrix $(\lambda I - A)$ is singular. Hence, by part (e) of the same theorem, $\det(\lambda I - A) = 0$. Since each step of this argument can be reversed, we have established the following theorem.

Theorem 5.2.3 λ is an eigenvalue of A if and only if

$$\det(\lambda I - A) = 0. \quad (5.2.4)$$

As we mentioned previously, (5.2.4) is called the *characteristic equation* of A . Because of the form of the equation, determinants will play a prominent role in this section. Although it is a useful theoretical device, the determinant is of little value for actual computations. Consequently determinants will appear only rarely in subsequent sections of the book.

It is not hard to see that $\det(\lambda I - A)$ is a polynomial in λ of degree n . It is called the *characteristic polynomial* of A .

Exercise 5.2.5

- (a) Show that the characteristic polynomial of

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

is $\lambda^2 - 5\lambda - 2$, a polynomial of degree 2.

- (b) Show that the characteristic polynomial of

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

is $\lambda^3 - 5\lambda^2 + \lambda + 1$, a polynomial of degree 3.

□

Exercise 5.2.6 Let

$$B = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}.$$

- (a) Show that the characteristic polynomial of B is $(\lambda - 1)(\lambda - 2)$ and the eigenvalues are 1 and 2. Thus B has two distinct eigenvalues.
- (b) Show that the characteristic polynomial of C is $(\lambda - 1)(\lambda - 1)$ and the eigenvalues are 1 and 1. Thus C does not have two distinct eigenvalues. It does have two eigenvalues if you count the eigenvalue 1 twice. The eigenvalue 1 is said to have *algebraic multiplicity* two, since it appears twice in the factorization of the characteristic polynomial of C .

□

Every n th degree polynomial equation has n complex roots, so A has n eigenvalues, some of which may be repeated. If A is a real matrix, the characteristic polynomial has real coefficients. But the zeros of a real polynomial are not necessarily real, so a real matrix can have complex, non-real, eigenvalues. This is why it makes sense to work with complex numbers from the beginning. Notice, however, that if a real matrix has complex eigenvalues, they must occur in complex conjugate pairs; that is, if $\lambda = \alpha + \beta i$ ($\alpha, \beta \in \mathbb{R}$) is an eigenvalue of $A \in \mathbb{R}^{n \times n}$, then so is the complex conjugate $\bar{\lambda} = \alpha - \beta i$.

Exercise 5.2.7 Prove the above assertion in two different ways:

- (a) First verify the following two basic facts about complex conjugates. If $z_1 = \alpha_1 + \beta_1 i \in \mathbb{C}$ and $z_2 = \alpha_2 + \beta_2 i \in \mathbb{C}$, then $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ and $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$.
- (b) Let $A \in \mathbb{R}^{n \times n}$. Let λ be a complex eigenvalue of A and let v be an associated eigenvector. Let $\bar{v}$ denote the vector obtained by taking complex conjugates of

the entries of v . Show that $\bar{v}$ is an eigenvector of A with associated eigenvalue $\bar{\lambda}$.

- (c) Let $p(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \cdots + a_n\lambda^n$ be a polynomial with real coefficients $a_0, \dots, a_n$. Show that for any $z = \alpha + \beta i \in \mathbb{C}$, $p(\bar{z}) = \overline{p(z)}$. Conclude that $p(z) = 0$ if and only if $p(\bar{z}) = 0$.

□

Now is a good time to mention one other important point about real matrices. Let $A \in \mathbb{R}^{n \times n}$ and suppose it has a real eigenvalue λ . Then in the homogeneous equation $(\lambda I - A)v = 0$, the coefficient matrix $\lambda I - A$ is real. The fact that $\lambda I - A$ is singular implies that the equation $(\lambda I - A)v = 0$ has nontrivial *real* solutions. We conclude that every real eigenvalue of a real matrix has a real eigenvector associated with it.

Exercise 5.2.8 Let $A \in \mathbb{C}^{n \times n}$.

- (a) Use the characteristic equation to show that A and A^T have the same eigenvalues.
- (b) Show that λ is an eigenvalue of A if and only if $\bar{\lambda}$ is an eigenvalue of A^* . (A^* denotes the complex conjugate transpose of A .)

□

There are some types of matrices for which the eigenvalues are obvious. The following theorem will prove very useful later on.

Theorem 5.2.9 Let $T \in \mathbb{C}^{n \times n}$ be a (lower- or upper-) triangular matrix. Then the eigenvalues of T are the main-diagonal entries $t_{11}, \dots, t_{nn}$.

Proof. The determinant of a triangular matrix R is $r_{11}r_{22} \cdots r_{nn}$, the product of the main-diagonal entries. Since T is triangular, so is $\lambda I - T$. Therefore the characteristic equation of T is $0 = \det(\lambda I - T) = (\lambda - t_{11})(\lambda - t_{22}) \cdots (\lambda - t_{nn})$, whose roots are obviously $t_{11}, \dots, t_{nn}$.

For a second proof see Exercise 5.2.19.

□

The following generalization of Theorem 5.2.9 is also quite useful.

Theorem 5.2.10 Let A be a block triangular matrix, say

$$A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1m} \\ & A_{22} & \cdots & A_{2m} \\ & & \ddots & \vdots \\ & & & A_{mm} \end{bmatrix}.$$

Then the set of eigenvalues of A , counting algebraic multiplicity, equals the union of the sets of eigenvalues of $A_{11}, A_{22}, \dots, A_{mm}$.

Proof. Because $\lambda I - A$ is block triangular, $\det(\lambda I - A) = \det(\lambda I - A_{11}) \det(\lambda I - A_{22}) \cdots \det(\lambda I - A_{mm})$. Thus the set of roots of the characteristic polynomial of A equals the union of the roots of the characteristic polynomials for $A_{11}, A_{22}, \dots, A_{mm}$. $\square$

Although A has been depicted as block upper triangular, the result is obviously valid for block lower-triangular matrices as well.

A second proof, which does not, however, yield the multiplicity information, is outlined in Exercise 5.2.20.

Theorem 5.2.11 *Let $v_1, \dots, v_k$ be eigenvectors of A associated with distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Then $v_1, \dots, v_k$ are linearly independent.*

Proof. See any text on linear algebra, or work through Exercise 5.2.21. $\square$

Recall from the previous section that a matrix $A \in \mathbb{C}^{n \times n}$ is *semisimple* or *nondefective* if there exists a set of n linearly independent eigenvectors of A . A matrix that is not semisimple is called *defective*.

Corollary 5.2.12 *If $A \in \mathbb{C}^{n \times n}$ has n distinct eigenvalues, then A is semisimple.*

If we pick a matrix “at random,” its characteristic equation is almost certain to have distinct roots. After all, the occurrence of repeated roots is an exceptional event. Thus a matrix chosen “at random” is almost certain to be semisimple. This is the basis of the claim made in the previous section that “most” matrices are semisimple. In addition it is not hard to show that every defective matrix has semisimple matrices arbitrarily close to it. Given a defective matrix (which must necessarily have some repeated eigenvalues), a slight perturbation suffices to split the repeated eigenvalues and yield a semisimple matrix (Exercise 5.4.32). In the language of topology, the set of semisimple matrices is *dense* in the set of all matrices.

A defective matrix must have at least one repeated eigenvalue. However, the presence of repeated eigenvalues does not guarantee that the matrix is defective.

Exercise 5.2.13 Find all eigenvalues and eigenvectors of the identity matrix $I \in \mathbb{C}^{n \times n}$. Conclude that (if $n > 1$) the identity matrix has repeated eigenvalues but is not defective. $\square$

So far we do not even know that defective matrices exist. The next exercise takes care of that.

Exercise 5.2.14

- The matrix $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has only the eigenvalue zero. (Why?) Show that it does not have two linearly independent eigenvectors.
- Find a 2×2 matrix that has only the eigenvalue zero but does have two linearly independent eigenvectors. (There is only one such matrix.)

(c) Show that the matrix

$$\begin{bmatrix} c & 1 & & & \\ & c & 1 & & \\ & & \ddots & \ddots & \\ & & & c & 1 \\ & & & & c \end{bmatrix}$$

is defective. A matrix of this form is called a *Jordan block*.

(d) Show that the matrix

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & 7 & x \\ 0 & 0 & 7 \end{bmatrix}$$

is defective if $x \neq 0$ and semisimple if $x = 0$.

□

Equivalence of Eigenvalue Problems and Polynomial Equations

We have seen that finding the eigenvalues of A is equivalent to finding the roots of the polynomial equation $\det(\lambda I - A) = 0$. Therefore, if we have a way of finding the roots of an arbitrary polynomial equation, then in principle we can find the eigenvalues of an arbitrary matrix. The next theorem shows that the converse is true as well: If we have a way of finding the eigenvalues of an arbitrary matrix, then in principle we can find the zeros of an arbitrary polynomial.

An arbitrary polynomial of degree n has the form $q(\lambda) = b_0 + b_1\lambda + b_2\lambda^2 + \cdots + b_n\lambda^n$, with $b_n \neq 0$. If we divide all coefficients of $q(\lambda)$ by b_n , we obtain a new polynomial $p(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \cdots + 1\lambda^n$ ($a_k = b_k/b_n$) that has same zeros and is *monic*; that is, its leading coefficient is $a_n = 1$. Since we can always replace the polynomial equation $q(\lambda) = 0$ by the equivalent equation $p(\lambda) = 0$, it suffices to consider monic polynomials. Given a monic polynomial, we define the *companion matrix* of p to be the $n \times n$ matrix

$$A = \begin{bmatrix} -a_{n-1} & -a_{n-2} & -a_{n-3} & \cdots & -a_0 \\ 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \end{bmatrix}. \quad (5.2.15)$$

This matrix has 1's on the subdiagonal, the opposites of the coefficients of p in the first row, and zeros elsewhere.

Theorem 5.2.16 Let $p(\lambda) = a_0 + a_1\lambda + a_2\lambda^2 + \cdots + 1\lambda^n$ be a monic polynomial of degree n , and let A be its companion matrix (5.2.15). Then the characteristic

polynomial of A is p . Thus the roots of the equation $p(\lambda) = 0$ are the eigenvalues of A .²

Proof. See Exercise 5.2.23. □

Necessity of Iterative Methods

Thanks to Theorem 5.2.16 we see that the eigenvalue problem and the problem of finding roots of a polynomial equation are equivalent. Although the algorithms that we will develop make no direct use of this equivalence, it does have an important theoretical implication. The polynomial root-finding problem is an old one that has attracted the interest of many great minds. In particular, early in the nineteenth century, Neils Henrik Abel was able to prove that there is no general formula³ for the roots of a polynomial equation of degree n if $n > 4$. (See, e.g., [37].) It follows that there is no general formula for the eigenvalues of an $n \times n$ matrix if $n > 4$.

Numerical methods can be divided into two broad categories—direct and iterative. A *direct method* is one that produces the result in a prescribed, finite number of steps. All versions of Gaussian elimination for solving $Ax = b$ are direct methods, and so are all of the methods that we developed in Chapter 3 for solving the least squares problem. By contrast an *iterative method* is one that produces a sequence of approximants that (hopefully) converges to the true solution of the problem. Each step or *iteration* of the algorithm produces a new approximant. In principle the sequence of approximants is infinite, but in practice we cannot run the algorithm forever. We stop once we have produced an approximant that is sufficiently good that we are willing to accept it as the solution. The number of iterations that will be needed to reach this point is usually not known in advance, although it can often be estimated.

Abel's theorem shows there are no direct methods for solving the general eigenvalue problem, for the existence of a finite, prespecified procedure would imply the existence of a (perhaps very complicated) formula for the solutions of an arbitrary polynomial equation. Therefore all eigenvalue algorithms are iterative.

Convergence Issues

In the next section we will study some of the simplest iterative methods. Each of these produces a sequence of vectors $q_1, q_2, q_3, \dots$ that (usually) converges to an eigenvector v as $j \rightarrow \infty$; that is, $q_j \rightarrow v$ as $j \rightarrow \infty$. For each iterative method that we introduce, we must concern ourselves with whether or not (or under what

²MATLAB has a command called `roots` that makes use of this correspondence. If x is a vector containing the coefficients of the polynomial p , the command `roots(x)` returns the roots of the equation $p(\lambda) = 0$. MATLAB does it as follows. The coefficients are normalized to make the polynomial monic. Then the companion matrix is formed. Finally the eigenvalues of the companion matrix are computed by the *QR* algorithm. These are the desired roots.

³We allow formulas that involve addition, subtraction, multiplication, division, and the extraction of roots.

conditions) it really does converge to an eigenvector. First of all, what do we mean when we write $q_j \rightarrow v$? We mean that each component of q_j converges to the corresponding component of v as $j \rightarrow \infty$. You can easily check that this componentwise convergence holds if and only if $\|q_j - v\|_\infty \rightarrow 0$ or $\|q_j - v\|_1 \rightarrow 0$ or $\|q_j - v\|_2 \rightarrow 0$, as $j \rightarrow \infty$.⁴ We will make extensive use of norms in our discussion of the convergence of iterative methods. In principle it does not matter which norm we use, as all norms on $\mathbb{C}^n$ are equivalent. This is so because $\mathbb{C}^n$ has finite dimension; see [70, Theorem II.3.1] or any other text on functional analysis.

Although the yes or no question of convergence is important, the *rate* of convergence is even more important. A convergent iterative method is of no practical interest if it takes millions of iterations to produce a useful result. Convergence rates will be discussed in the coming section in connection with the iterative methods introduced there.

A Variety of Iterative Methods

As we have already indicated, the methods to be introduced in the next section generate sequences of vectors. In subsequent sections of this chapter we will develop more sophisticated methods that generate sequences of matrices $A_1, A_2, A_3, \dots$, such that each matrix in the sequence has the same eigenvalues as the original matrix A , and the matrices converge to a simple form such as upper-triangular form, from which the eigenvalues can be read. In Chapter 6 we will introduce yet another class of iterative methods that generate sequences of subspaces of $\mathbb{C}^n$. By the middle of Chapter 6 it will be clear that all of these methods, whether they generate vectors, matrices, or subspaces, are intimately related.

Additional Exercises

Exercise 5.2.17 Let $A \in \mathbb{R}^{n \times m}$ have SVD $A = U\Sigma V^T$. The right singular vectors are $v_1, \dots, v_m$, the columns of V ; the left singular vectors are $u_1, \dots, u_n$, the columns of U . Show that $v_1, \dots, v_m$ are linearly independent eigenvectors of $A^T A$, and $u_1, \dots, u_n$ are linearly independent eigenvectors of AA^T , associated with eigenvalues $\sigma_1^2, \dots, \sigma_r^2, 0, 0, \dots$, where r is the rank of A . $\square$

Exercise 5.2.18 Let λ be an eigenvalue of A , and consider the set

$$\mathcal{S}_\lambda = \{v \in \mathbb{C}^n \mid Av = \lambda v\},$$

which consists of all eigenvectors associated with λ , together with the zero vector.

(a) Show that if $v_1 \in \mathcal{S}_\lambda$ and $v_2 \in \mathcal{S}_\lambda$, then $v_1 + v_2 \in \mathcal{S}_\lambda$.

(b) Show that if $v \in \mathcal{S}_\lambda$ and $c \in \mathbb{C}$, then $cv \in \mathcal{S}_\lambda$.

⁴Our discussion of vector norms in Section 2.1 was restricted to norms on $\mathbb{R}^n$. However, that entire discussion can be carried over to the complex setting.

Thus $\mathcal{S}_\lambda$ is a subspace of $\mathbb{C}^n$. It is called the *eigenspace* of A associated with λ . Even if λ is not an eigenvalue, we can still define the space $\mathcal{S}_\lambda$, but in this case $\mathcal{S}_\lambda = \{0\}$. Clearly λ is an eigenvalue of A if and only if $\mathcal{S}_\lambda \neq \{0\}$. $\square$

Exercise 5.2.19 This exercise gives a second proof that the main-diagonal entries of a triangular matrix are its eigenvalues. Let T be a triangular matrix. For the sake of argument let us assume that it is upper triangular. Let μ be a complex number.

- (a) Suppose $\mu = t_{ii}$ for some i . We will show that μ is an eigenvalue by producing an eigenvector associated with μ . That is, we will produce a nonzero $v \in \mathbb{C}^n$ such that $(\mu I - T)v = 0$. It may be the case that $\mu = t_{ii}$ for several different i . Let j be the smallest index for which $\mu = t_{jj}$. Thus $\mu \neq t_{ii}$ for all $i < j$. Show that if $j = 1$, then $(\mu I - T)e_1 = 0$, where $e_1 = [1, 0, \dots, 0]^T$. If $j \neq 1$, partition T as follows:

$$T = \begin{bmatrix} T_{11} & x & T_{13} \\ 0 & t_{jj} & y^T \\ 0 & 0 & T_{13} \end{bmatrix},$$

where T_{11} is $(j-1) \times (j-1)$, $x \in \mathbb{C}^{j-1}$, and so on. Prove that there is a unique vector v of the form

$$v = \begin{bmatrix} z \\ 1 \\ 0 \end{bmatrix},$$

where $z \in \mathbb{C}^{j-1}$, such that $(\mu I - T)v = 0$. Thus μ is an eigenvalue of T .

- (b) Show that if $\mu \neq t_{ii}$ for all i , and $v \in \mathbb{C}^n$ satisfies $(\mu I - T)v = 0$, then $v = 0$. (Do not invoke determinants; just prove the result directly from the equation.) Thus μ is not an eigenvalue of T .

$\square$

Exercise 5.2.20 Let $A \in \mathbb{C}^{n \times n}$ be a block triangular matrix:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix},$$

where $A_{11} \in \mathbb{C}^{j \times j}$ and $A_{22} \in \mathbb{C}^{k \times k}$, $j + k = n$.

- (a) Let λ be an eigenvalue of A_{11} with eigenvector v . Show that there exists a $w \in \mathbb{C}^k$ such that $\begin{bmatrix} v \\ w \end{bmatrix}$ is an eigenvector of A with associated eigenvalue λ .
- (b) Let λ be an eigenvalue of A_{22} with associated eigenvector w , and suppose λ is not also an eigenvalue of A_{11} . Show that there is a unique $v \in \mathbb{C}^j$ such that $\begin{bmatrix} v \\ w \end{bmatrix}$ is an eigenvector of A with associated eigenvalue λ .

- (c) Let λ be an eigenvalue of A with associated eigenvector $\begin{bmatrix} v \\ w \end{bmatrix}$. Show that either w is an eigenvector of A_{22} with associated eigenvalue λ or v is an eigenvector of A_{11} with associated eigenvalue λ . (Hint: Either $w \neq 0$ or $w = 0$. Consider these two cases.)
- (d) Combining parts (a), (b), and (c), show that λ is an eigenvalue of A if and only if it is an eigenvalue of A_{11} or A_{22} .

□

Exercise 5.2.21 Prove Theorem 5.2.11 by induction on k . You might like to make use of the following outline. First show that the theorem is true for $k = 1$. Then suppose k is arbitrary and show that if the theorem holds for sets of $k - 1$ vectors (induction hypothesis), then it must also hold for sets of k vectors. Suppose

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0. \quad (5.2.22)$$

To establish independence of $v_1, \dots, v_k$, you need to show that $c_1 = c_2 = \cdots = c_k = 0$. Apply $(A - \lambda_k I)$ to both sides of (5.2.22). Simplify the result by using the equations $Av_j = \lambda_j v_j$ for $j = 1, \dots, k$, and note that one term is eliminated from the left-hand side. Use the induction hypothesis to conclude that $c_1 = c_2 = \cdots = c_{k-1} = 0$. Point out clearly how you are using the distinctness of the eigenvalues here. Finally, go back to (5.2.22) and show that $c_k = 0$.

□

Exercise 5.2.23 Parts (a) and (b) of this exercise provide two independent proofs of Theorem 5.2.16. Let $p(\lambda) = a_0 + a_1 \lambda + a_2 \lambda^2 + \cdots + 1 \lambda^n$, and let A be its companion matrix (5.2.15).

- (a) Let $\lambda \in \mathbb{C}$ be an eigenvalue of A with associated eigenvector v . Write down and examine the equation $Av = \lambda v$. Demonstrate that v must be a multiple of the vector

$$\begin{bmatrix} \lambda^{n-1} \\ \vdots \\ \lambda^2 \\ \lambda \\ 1 \end{bmatrix}, \quad (5.2.24)$$

and λ must satisfy $p(\lambda) = 0$. Conversely, show that if $p(\lambda) = 0$, then λ is an eigenvalue of A with eigenvector v given by (5.2.24). Thus the eigenvalues of A are exactly the zeros of p .

- (b) For $j = 1, \dots, n$, let $\chi_j(\lambda) = \lambda^j + a_{n-1} \lambda^{j-1} + \cdots + a_{n-j+1} \lambda + a_{n-j}$. Thus χ_j is a monic polynomial of degree j , and $\chi_n = p$. Let A_j be the companion matrix of χ_j . Prove by induction on j that $\det(\lambda I - A_j) = \chi_j(\lambda)$, $j = 1, \dots, n$. Hint: Expand the last column of the determinant $\det(\lambda I - A_j)$ to obtain the recurrence

$$\det(\lambda I - A_j) = \lambda \det(\lambda I - A_{j-1}) + a_{n-j}.$$

□

5.3 THE POWER METHOD AND SOME SIMPLE EXTENSIONS

Let $A \in \mathbb{C}^{n \times n}$. To avoid complicating the analysis we assume that A is semisimple. This means that A has a set of n linearly independent eigenvectors $v_1, \dots, v_n$, which must then form a basis for $\mathbb{C}^n$. Let $\lambda_1, \dots, \lambda_n$ denote the eigenvalues associated with $v_1, \dots, v_n$, respectively. Let us assume that the vectors are ordered so that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$. If $|\lambda_1| > |\lambda_2|$, λ_1 is called the *dominant eigenvalue* and v_1 is called a *dominant eigenvector* of A .

If A has a dominant eigenvalue, then we can find it and an associated dominant eigenvector by the *power method*. The basic idea is to pick a vector q and form the sequence

$$q, Aq, A^2q, A^3q, \dots$$

To calculate this sequence it is not necessary to form the powers of A explicitly. Each vector in the sequence can be obtained by multiplying the previous vector by A , e.g. $A^{j+1}q = A(A^j q)$. It is easy to show that the sequence converges, in a sense, to a dominant eigenvector, for almost all choices of q . Since $v_1, \dots, v_n$ form a basis for $\mathbb{C}^n$, there exist constants $c_1, \dots, c_n$ such that

$$q = c_1 v_1 + c_2 v_2 + \dots + c_n v_n.$$

So far we do not know $v_1, \dots, v_n$, so we do not know what $c_1, \dots, c_n$ are either. However, it is clear that for practically any choice of q , c_1 will be nonzero (Exercise 5.3.43). The argument that follows is valid for every q for which $c_1 \neq 0$. Multiplying q by A , we have

$$\begin{aligned} Aq &= c_1 A v_1 + c_2 A v_2 + \dots + c_n A v_n \\ &= c_1 \lambda_1 v_1 + c_2 \lambda_2 v_2 + \dots + c_n \lambda_n v_n. \end{aligned}$$

Similarly

$$A^2 q = c_1 \lambda_1^2 v_1 + c_2 \lambda_2^2 v_2 + \dots + c_n \lambda_n^2 v_n,$$

and in general

$$A^j q = c_1 \lambda_1^j v_1 + c_2 \lambda_2^j v_2 + \dots + c_n \lambda_n^j v_n. \quad (5.3.1)$$

Since λ_1 dominates the other eigenvalues, the component of $A^j q$ in the direction of v_1 becomes steadily greater relative to the other components as j increases. This can be made clearer by rewriting (5.3.1) as

$$A^j q = \lambda_1^j (c_1 v_1 + c_2 (\lambda_2/\lambda_1)^j v_2 + \dots + c_n (\lambda_n/\lambda_1)^j v_n). \quad (5.3.2)$$

The fact that every multiple of an eigenvector is also an eigenvector means that the magnitude of an eigenvector is unimportant; only the direction matters. Therefore the factor λ_1^j in (5.3.2) is, in principle, unimportant. Thus, instead of the sequence $(A^j q)$, let us consider the rescaled sequence (q_j) , where $q_j = A^j q / \lambda_1^j$. From (5.3.2) it is clear that $q_j \rightarrow c_1 v_1$ as $j \rightarrow \infty$. Indeed, for any vector norm,

$$\begin{aligned} \|q_j - c_1 v_1\| &= \|c_2 (\lambda_2/\lambda_1)^j v_2 + \dots + c_n (\lambda_n/\lambda_1)^j v_n\| \\ &\leq |c_2| |\lambda_2/\lambda_1|^j \|v_2\| + \dots + |c_n| |\lambda_n/\lambda_1|^j \|v_n\| \\ &\leq (|c_2| \|v_2\| + \dots + |c_n| \|v_n\|) |\lambda_2/\lambda_1|^j. \end{aligned}$$

Here we have used the fact that $|\lambda_i| \leq |\lambda_2|$ for $i = 3, \dots, n$. Letting $C = |c_2| \|v_2\| + \dots + |c_n| \|v_n\|$, we have

$$\|q_j - c_1 v_1\| \leq C |\lambda_2/\lambda_1|^j \quad j = 1, 2, 2, \dots \quad (5.3.3)$$

Since $|\lambda_1| > |\lambda_2|$, it follows that $|\lambda_2/\lambda_1| \rightarrow 0$ as $j \rightarrow \infty$. Thus $\|q_j - c_1 v_1\| \rightarrow 0$. This means that for sufficiently large j , q_j is a good approximation to the dominant eigenvector $c_1 v_1$. The number $\|q_j - c_1 v_1\|$ gives a measure of the error in this approximation. From (5.3.3) we see that, roughly speaking, the magnitude of the error is decreased by a factor $|\lambda_2/\lambda_1|$ with each iteration. Therefore the ratio $|\lambda_2/\lambda_1|$ is an important indicator of the rate of convergence.

The convergence behavior exhibited by the power method is called linear convergence. In general a sequence (x_j) that converges to x is said to converge *linearly* if there is a number r satisfying $0 < r < 1$, such that

$$\lim_{j \rightarrow \infty} \frac{\|x_{j+1} - x\|}{\|x_j - x\|} = r. \quad (5.3.4)$$

This means that $\|x_{j+1} - x\| \approx r \|x_j - x\|$ for sufficiently large j .⁵ The number r is called the *convergence ratio* or *contraction number* of the sequence. The power method generally exhibits linear convergence with convergence ratio $r = |\lambda_2/\lambda_1|$ (Exercise 5.3.44).

In practice the sequence $q_j = A^j q / \lambda_1^j$ is inaccessible because we do not know λ_1 in advance. On the other hand, it is impractical to work with $A^j q$ itself because $\|A^j q\| \rightarrow \infty$ if $|\lambda_1| > 1$ and $\|A^j q\| \rightarrow 0$ if $|\lambda_1| < 1$. In order to avoid overflow or underflow and to recognize when the iterates are converging, we must employ some kind of scaling. Thus we let $q_0 = q$ and define

$$q_{j+1} = A q_j / \sigma_{j+1},$$

where σ_{j+1} is some convenient scaling factor. The exact choice of scale factor is unimportant, since we are interested in the directions of the vectors, not the lengths. A simple and convenient strategy is to take σ_{j+1} to be that entry of $A q_j$ that is largest in absolute value. The effect of this choice is that the largest component of each q_j is 1, and the sequence converges to a dominant eigenvector whose largest component is 1.

Iterations of the power method are relatively inexpensive. The cost of multiplying the $n \times n$ matrix A by q_j is $2n^2$ flops. The normalizing operations require only $O(n)$ work, as you can easily verify, so the total cost of a power iteration is about $2n^2$ flops. Thus m iterations will cost $2n^2 m$ flops. This count assumes that A is not a sparse matrix. If A is sparse, the cost of calculating $A q_j$ will be considerably less than $2n^2$ flops. For example, if A has five nonzero entries in each row, the cost of computing $A q_j$ is only about $10n$ flops.

⁵Some authors also allow $r = 1$, but it seems preferable to think of that case as representing *slower than linear* convergence. A sequence that satisfies (5.3.4) with $r = 1$ converges very slowly indeed.

Our development of the power method has assumed that the matrix is semisimple. This turns out not to be crucial; essentially the same conclusions can be drawn for defective matrices.

Example 5.3.5 We will use the power method to calculate a dominant eigenvector of

$$A = \begin{bmatrix} 9 & 1 \\ 1 & 2 \end{bmatrix}.$$

If we start with the vector $q_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$, then on the first step we have $Aq_0 = \begin{bmatrix} 10 & 3 \end{bmatrix}^T$. Dividing by the scale factor $\sigma_1 = 10$, we get $q_1 = \begin{bmatrix} 1 & 0.3 \end{bmatrix}^T$. Then $Aq_1 = \begin{bmatrix} 9.3 & 1.6 \end{bmatrix}^T$, $\sigma_2 = 9.3$, and $q_2 = \begin{bmatrix} 1 & 0.172034 \end{bmatrix}^T$. Subsequent iterates are listed in Table 5.1. Only the second component of each q_j is listed because the

j	σ_j	q_j (second component)
3	9.172043	0.146541
4	9.146541	0.141374
5	9.141374	0.140323
6	9.140323	0.140110
7	9.140110	0.140066
8	9.140066	0.140057
9	9.140057	0.140055
10	9.140055	0.140055

Table 5.1 Iterates of power method

first component is always 1. We see that after 10 iterations the sequence of vectors (q_j) has converged to six decimal places. Thus (to six decimal places)

$$v_1 = \begin{bmatrix} 1.0 \\ 0.140055 \end{bmatrix}.$$

The sequence σ_j converges to the dominant eigenvalue $\lambda_1 = 9.140055$. $\square$

In Example 5.3.5 and also in Exercises 5.3.7, 5.3.10, and 5.3.11, the iterates converge reasonably rapidly. This is so because in each case the ratio $|\lambda_2/\lambda_1|$ is fairly small. For most matrices this ratio is not so favorable. Not uncommonly $|\lambda_2/\lambda_1|$ is very close to 1, in which case convergence is slow. See Exercise 5.3.9.

Exercise 5.3.6 Calculate the characteristic polynomial of the matrix of Example 5.3.5, and use the quadratic formula to find the eigenvalues. Calculate the dominant eigenvector as well, and verify that the results of Example 5.3.5 are correct. Calculate the ratios of errors $\|q_{j+1} - v_1\|/\|q_j - v_1\|$, $j = 0, 1, 2, \dots$, and note that they agree very well with the theoretical convergence ratio $|\lambda_2/\lambda_1|$. Why is the agreement so good? Would you expect the agreement to be as good if A were 3×3 or larger? $\square$

Exercise 5.3.7 Let $A = \begin{bmatrix} 8 & 1 \\ -2 & 1 \end{bmatrix}$.

- Use the power method with $q_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ to calculate the dominant eigenvalue and eigenvector of A . Tabulate q_j and σ_j at each step. Iterate until the q_j and σ_j have converged to at least six decimal places.
- Now that you have calculated the eigenvector v , calculate the ratios

$$\|q_{j+1} - v\| / \|q_j - v\|, \quad j = 0, 1, 2, \dots$$

to find the observed rate of convergence. Using the characteristic equation, solve for the eigenvalues and calculate the ratio $|\lambda_2/\lambda_1|$, which gives the theoretical rate of convergence. How well does theory agree with practice in this case?

□

Exercise 5.3.8 Work this problem by hand. Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Carry out the power method

with starting vector $q_0 = \begin{bmatrix} a & b \end{bmatrix}^T$, where $a \geq 0$, $b \geq 0$, and $a \neq b$. Explain why the sequence fails to converge. Why does the convergence argument given above fail in this case?

□

Exercise 5.3.9 Work this problem by hand. Only at the very last step will you need a calculator.

Let $A = \begin{bmatrix} 0.99 & 0 \\ 0 & 1 \end{bmatrix}$.

- Find the eigenvalues of A and associated eigenvectors.
- Carry out power iterations starting with $q_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$. Derive a general expression for q_j .
- How many iterations are required in order to obtain $\|q_j - v_1\|_\infty / \|v_1\|_\infty < 10^{-6}$?

□

Exercise 5.3.10 Let $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 9 & 2 \\ 0 & -1 & 2 \end{bmatrix}$.

- Using MATLAB, apply the power method to A , starting with

$$q_0 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T.$$

Do at least ten iterations. Here are some sample MATLAB commands.

```
A = [ 1 1 1 ; -1 9 2 ; 0 -1 2 ]
q = ones(3,1)
```



```

iterate(:,1) = q;
for j = 1:10
    q = A*q;
    [bgst,index] = max(abs(q));
    sigma(j+1) = q(index(1));
    q = q/sigma(j+1);
    iterate(:,j+1) = q;
end
iterate
sigma

```

If you store some of these (or similar) commands in a file named, say, `power.m`, then each time you type the command `power`, MATLAB will execute that sequence of instructions. The numbers are stored to much greater accuracy than what is normally displayed on the screen. If you wish to display more digits, type `format long`.

- (b) Use the command `[V,D] = eig(A)` to get the eigenvalues and eigenvectors of A . Compare the dominant eigenvalue from D with the scale factors that you computed in part (a).
- (c) Locate a dominant eigenvector from V and rescale it appropriately, so that you can compare it with the iterates from part (a). Calculate the ratios

$$\|q_{j+1} - v\|_2 / \|q_j - v\|_2 \quad \text{for } j = 1, 2, 3, \dots$$

- (d) Using D from part (b), calculate the ratio $|\lambda_2/\lambda_1|$ and compare it to the ratios computed in part (b).

□

Exercise 5.3.11 Repeat Exercise 5.3.10 using the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 9 & 2 \\ -4 & -1 & 2 \end{bmatrix}.$$

What is different here? The ratios of errors are more erratic. Compute

$$\sqrt[j]{\|q_{j+1} - v\| / \|q_1 - v\|}, \quad j = 1, 2, 3, \dots,$$

and compare these values with $|\lambda_2/\lambda_1|$. Why would you expect these to be close?

□

Exercise 5.3.12 Repeat Exercise 5.3.10 using the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 3 & 2 \\ -4 & -1 & 2 \end{bmatrix}.$$

What happened this time?

□

Inverse Iteration and the Shift-and-Invert Strategy

We continue to assume that $A \in \mathbb{C}^{n \times n}$ is semisimple with linearly independent eigenvectors $v_1, \dots, v_n$ and associated eigenvalues $\lambda_1, \dots, \lambda_n$, arranged in order of descending magnitude. If A is nonsingular, we can apply the power method to A^{-1} . This is the *inverse power method* or *inverse iteration*. (By contrast, the power method applied to A is sometimes called *direct iteration*.)

Exercise 5.3.13 Suppose $A \in \mathbb{C}^{n \times n}$ is nonsingular. Then all eigenvalues of A are nonzero. Show that if v is an eigenvector of A with associated eigenvalue λ , then v is also an eigenvector of A^{-1} with associated eigenvalue λ^{-1} . $\square$

From Exercise 5.3.13 we see that A^{-1} has linearly independent eigenvectors $v_n, v_{n-1}, \dots, v_1$ with eigenvalues $\lambda_n^{-1}, \lambda_{n-1}^{-1}, \dots, \lambda_1^{-1}$. If $|\lambda_n|^{-1} > |\lambda_{n-1}^{-1}|$ (i.e. $|\lambda_{n-1}| > |\lambda_n|$) and we start with a vector $q = c_n v_n + \dots + c_1 v_1$ for which $c_n \neq 0$, then the inverse power iterates will converge to a multiple of v_n , an eigenvector associated with the smallest eigenvalue of A . The convergence ratio is $|\lambda_{n-1}^{-1}/\lambda_n^{-1}| = |\lambda_n/\lambda_{n-1}|$, so convergence will be fast when $|\lambda_{n-1}| \gg |\lambda_n|$. This suggests that we *shift* the eigenvalues so that the smallest one is very close to zero. To understand how this works, we need only the following simple result.

Exercise 5.3.14 Let $A \in \mathbb{C}^{n \times n}$, and let $\rho \in \mathbb{C}$. Show that if v is an eigenvector of A with eigenvalue λ , then v is also an eigenvector of $A - \rho I$ with eigenvalue $\lambda - \rho$. $\square$

From Exercise 5.3.14 we see that if A has eigenvalues $\lambda_1, \dots, \lambda_n$, then $A - \rho I$ has eigenvalues $\lambda_1 - \rho, \lambda_2 - \rho, \dots, \lambda_n - \rho$. The scalar ρ is called a *shift*. If the shift is chosen so that it is a good approximation to λ_n , then $|\lambda_{n-1} - \rho| \gg |\lambda_n - \rho|$, and inverse iteration applied to $A - \rho I$ will converge rapidly to a multiple of v_n . Actually there is nothing special about λ_n ; the shift ρ can be chosen to approximate any one of the eigenvalues of A . If ρ is a good enough approximation to λ_i that $\lambda_i - \rho$ is much smaller than any other eigenvalue of $A - \rho I$, then (for almost any starting vector q) inverse iteration applied to $A - \rho I$ will converge to a multiple of the eigenvector v_i . The convergence ratio is $|(\lambda_i - \rho)/(\lambda_k - \rho)|$, where $\lambda_k - \rho$ is the second smallest eigenvalue of $A - \rho I$. the closer ρ is to λ_i , the swifter the convergence will be.

Shifts can be used in conjunction with direct iteration as well, but they are not nearly so effective in that context. The combination of shifting and inverse iteration works well because there exist shifts for which one eigenvalue of the shifted matrix is much smaller than all other eigenvalues. In contrast, there (usually) do not exist shifts for which one eigenvalue is much larger than all other eigenvalues.

To get rapid convergence, we shift first and then apply inverse iteration. This is called the *shift-and-invert* strategy. It is one of the most important ideas for eigenvalue computations.

Exercise 5.3.15 Work this problem by hand. Diagonal matrices such as

$$A = \begin{bmatrix} 2.99 & 0 & 0 \\ 0 & 1.99 & 0 \\ 0 & 0 & 1.00 \end{bmatrix}$$

have particularly simple eigensystems.

- Find the eigenvalues and eigenvectors of A .
- Find the eigenvalues of $A - \rho I$ and $(A - \rho I)^{-1}$, where $\rho = 0.99$. Perform both direct and inverse iteration on $A - \rho I$, starting with $q_0 = [1 \ 1 \ 1]^T$. To which eigenvector does each sequence converge? Which converges faster?
- Perform inverse iteration with values (i) $\rho = 2.00$ and (ii) $\rho = 3.00$, using the same starting vector. To which eigenvectors do these sequences converge?

□

Let us consider some of the practical aspects of inverse iteration. The iterates will satisfy

$$q_{j+1} = (A - \rho I)^{-1} q_j / \sigma_{j+1},$$

but there is no need to calculate $(A - \rho I)^{-1}$ explicitly. Instead one can solve the linear system $(A - \rho I)\hat{q}_{j+1} = q_j$ and then set $q_{j+1} = \hat{q}_{j+1}/\sigma_{j+1}$, where σ_{j+1} equals the component of $\hat{q}_{j+1}$ that is largest in magnitude. If the system is to be solved by Gaussian elimination, then the LU decomposition of $A - \rho I$ has to be done only once. Then each iteration consists of forward substitution, back substitution, and normalization. For a full $n \times n$ matrix the cost is $\frac{2}{3}n^3$ flops for the LU decomposition plus $2n^2$ flops per iteration.

Example 5.3.16 Let us apply inverse iteration with a shift to the matrix

$$A = \begin{bmatrix} 9 & 1 \\ 1 & 2 \end{bmatrix}.$$

We know from Example 5.3.5 that 9 is a good approximation to an eigenvalue of A . Even if we did not know this, we might expect as much from the dominant 9 in the $(1, 1)$ position of the matrix. Thus it is reasonable to use the shift $\rho = 9$. Starting with $q_0 = [1 \ 1]^T$, we solve the system $(A - 9I)\hat{q}_1 = q_0$ to get $\hat{q}_1 = [8.0 \ 1.0]^T$. We then rescale by taking $\sigma_1 = 8$ to get $q_1 = [1.0 \ 0.125]^T$. Solving $(A - 9I)\hat{q}_2 = q_1$, we get $\hat{q}_2 = [7.125 \ 1.0]^T$. Thus $\sigma_2 = 7.125$, and $q_2 = [1.0 \ 0.140351]^T$. Subsequent iterates are listed in Table 5.2. As in Table 5.1, we have listed only the second component of q_j because the first component is always 1. After five iterations we have the eigenvector $v_1 = [1.0 \ 0.140055]^T$ correct to six decimal places. The good choice of shift gave much faster convergence than we had in Example 5.3.5. The scale factors converge to an eigenvalue of

j	σ_j	q_j (second component)
3	7.140351	0.140449
4	7.140449	0.140055
5	7.140055	0.140055

Table 5.2 Iterates of inverse power method

$(A - 9I)^{-1}$: $(\lambda_1 - 9)^{-1} = 7.140055$. Solving for λ_1 , we obtain $\lambda_1 = 9.140055$. Another way to obtain the eigenvalue is to compute the Rayleigh quotient, which will be discussed below. $\square$

Exercise 5.3.17 Let $A = \begin{bmatrix} 8 & 1 \\ -2 & 1 \end{bmatrix}$.

- Use inverse iteration with $\rho = 8$ and $q_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ to calculate an eigenvalue and eigenvector of A . (On this small problem it may be easiest simply to calculate $B = (A - 8I)^{-1}$ and perform direct iterations with B .)
- Now that you have calculated an eigenvector v , calculate the ratios

$$\|q_{j+1} - v\| / \|q_j - v\|$$

for $j = 0, 1, 2, \dots$, to find the observed rate of convergence. Solve for the eigenvalues using the characteristic equation or MATLAB and calculate the theoretical convergence rate $|(\lambda_1 - 8)/(\lambda_2 - 8)|$. How well does theory agree with practice? $\square$

Exercise 5.3.18 Let $A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 9 & 2 \\ 0 & -1 & 2 \end{bmatrix}$.

- Using MATLAB, apply inverse iteration to A with shift $\rho = 9$, starting with $q_0 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$. For simplicity, just form $B = (A - 9I)^{-1}$ (`B = inv(A-9*eye(3))` in MATLAB) and iterate with B . Some of the commands shown in Exercise 5.3.10 may be useful here. Use `format long` to view more digits of your iterates. Do at least ten iterations.
- Use `[V,D] = eig(A)` to get the true dominant eigenvector. Calculate the errors $\|q_j - v\|_2$ and ratios $\|q_{j+1} - v\|_2 / \|q_j - v\|_2$ for $j = 1, 2, 3, \dots$. Compute the theoretical convergence rate (from the known eigenvalues) and compare it with these ratios. Notice that we have faster convergence than we had when we applied the power method to this same matrix in Exercise 5.3.10. $\square$

Exercise 5.3.19 The matrix `west0479` is a sparse 479×479 matrix that is supplied with MATLAB. We used this matrix previously in Exercises 1.6.3 and 1.9.2. Some useful commands are

```
load west0479
A = west0479; % Just to get a short abbreviation.
issparse(A) % = 1 for sparse, = 0 for dense matrices.
size(A)
nnz(A) % number of nonzeros of A
disp(A)
spy(A)
```

- (a) It is known that A has an eigenvalue near $\rho = 15 + 35i$. Use inverse iteration with shift ρ to calculate this eigenvalue. (Solution: $\lambda = 17.548546093831 + 34.237822957500i$.) More useful commands:

```
n = 479;
shift = 15 + 35i;
[L,U,P] = lu(A - shift*speye(n));
q = ones(n,1);
for j = 1:itnum
    q = P*q;
    q = L\q;
    q = U\q;
    [bgst,index] = max(abs(q));
    sigma(j+1) = q(index(1));
    q = q/sigma(j+1);
end
```

Add some commands of your own to obtain the desired output

- (b) Explain the function of the four commands involving L , U , and P in part (a). How do they help make the algorithm economical?
- (c) When the matrix is large and sparse, the speed of the algorithm depends heavily upon how it is implemented. Run three different implementations of inverse iteration, keeping track of the execution time:
- Use the implementation shown in part (a).
 - Let $B = A - \text{shift} \cdot \text{speye}(n)$ and use $q = B \backslash q$ to do the inverse iteration. This lazy implementation forces MATLAB to redo the sparse LU decomposition on each iteration.
 - Let $C = \text{full}(\text{inv}(B))$, where B is as before, and use $q = C * q$ to do the iteration. (Type `spy(C)` to verify that C is not at all sparse.)

Time, say, 20 iterations by each method. Also determine separately the time it takes to compute the sparse LU decomposition and the time it takes to compute the inverse. Discuss your results. More sample code:

```
t1 = cputime;
```

```
[L,U,P] = lu(A - shift*speye(n));
t2 = cputime;
decomp_time = t2 - t1
```

□

Exercise 5.3.20 This exercise is similar to the previous one, except that it features a scalable family of matrices. The MATLAB command `delsq` generates sparse matrices associated with discretized Laplacian (“del squared”) operators on various regions. For example, try

```
m = 32
A = delsq(numgrid('S',m));
issparse(A)
size(A)
```

This produces a 900×900 matrix. You can make a larger or smaller matrix by adjusting the size of m . In general the matrix A has dimension $(m - 2)^2$. We said more about this family in Exercise 1.6.4.

- (a) This matrix is symmetric and positive definite. Its largest eigenvalue is near 8. Use inverse iteration with shift $\rho = 8$ to estimate this eigenvalue to five decimal places accuracy. (The solution depends on m .) Borrow code from Exercise 5.3.19, part (a).
- (b) Repeat part (c) of Exercise 5.3.19 using a `delsq` matrix. Make m large enough that you get impressive results, but not so large that you die of old age before the results come back.

□

Inverse Iteration with Exact Shifts

The introduction of shifts allows us to find any eigenvector, not just those associated with the largest and smallest eigenvalues. However, in order to find a given eigenvector, we must have a good approximation to the associated eigenvalue to begin with. It follows that a good application for inverse iteration is to find eigenvectors associated with eigenvalues that have been computed by some other means. Using the computed eigenvalue as a shift, we typically obtain an excellent eigenvector in just one or two iterations. We call this *inverse iteration with an exact shift*. It is an important application of inverse iteration.

Exercise 5.3.21 An oracle tells you that $\lambda = 17.548546093831 + 34.237822957500i$ is an eigenvalue of the matrix `west0479` from MATLAB. Perform one step of inverse iteration, using this computed eigenvalue as a shift, to get an estimate v of an associated eigenvector. Borrow MATLAB code from Exercise 5.3.19. To get an idea of the quality of this estimated eigenvector, calculate the residual $\|Av - \lambda v\|_2 / \|v\|_2$.

A residual of zero would indicate a perfect eigenpair. If you work this problem correctly, you will get an impressively small residual. (*Warning:* A small residual does not absolutely guarantee an accurate eigenvector. See the discussion of sensitivity of eigenvalues and eigenvectors in Section 6.5.) $\square$

Inverse iteration with exact shifts works amazingly well when one considers the circumstances. If ρ is an eigenvalue of A , then $A - \rho I$ is singular, so $(A - \rho I)^{-1}$ does not exist. How can we possibly do inverse iteration with this shift? Well, in practice ρ is (almost) never exactly an eigenvalue; there is (almost) always some error. Therefore $A - \rho I$ is (almost) always nonsingular. Furthermore, recall from Chapter 2 that in numerical practice it is impossible to distinguish between singular and nonsingular matrices. Therefore, even if $A - \rho I$ is singular, we will (almost) never be able to detect the fact. Therefore the method (almost) never breaks down in practice.

However, if ρ is nearly an eigenvalue, this means that $A - \rho I$ is nearly singular; that is, it is ill conditioned (Exercise 5.3.47). One might reasonably fear that the ill condition of $A - \rho I$ would spoil the computation, since at each step a system of the form $(A - \rho I)\hat{q}_{j+1} = q_j$ must be solved. Fortunately the ill conditioning turns out not to be a problem in this context. Consider a single step $(A - \rho I)\hat{q}_1 = q_0$. If ρ is nearly an eigenvalue of A , then $\hat{q}_1$ is already very close to an eigenvector of A , if the system has been solved exactly. But suppose instead that the system was solved by Gaussian elimination with partial pivoting. Then we know from Chapter 2 (Theorem 2.7.14 and ensuing discussion) that the computed $\hat{q}_1$ actually satisfies a perturbed system $(A + \delta A - \rho I)\hat{q}_1 = q_0$, where δA is normally tiny relative to A . Since δA is tiny, ρ should also be a good approximation to an eigenvalue of $A + \delta A$, so $\hat{q}_1$ should be nearly an eigenvector of $A + \delta A$, which should then also be close to an eigenvector of A . Thus inverse iteration should be effective.

Whether or not all this is true depends upon whether the small perturbation in A really does cause only small perturbations of the eigenvalue and eigenvector. This depends in turn on condition numbers for the eigenvalue and eigenvector, which are independent of the condition number of $A - \rho I$. Thus the effectiveness of inverse iteration does not depend on the condition number of $A - \rho I$. We will discuss condition numbers for eigenvalues and eigenvectors in Section 6.5.

The Rayleigh Quotient

In using inverse iteration to calculate an eigenvector of A , there is no reason why one could not use a different shift at each step. This might be useful in situations in which the eigenvalue is not known in advance. If q_j is close enough to an eigenvector, it should somehow be possible to use q_j to obtain an estimate of the associated eigenvalue. This estimate could then be used as the shift for the next iteration. In this way, we would get an improved shift, hence an improved convergence ratio, for each iteration.

Suppose $q \in \mathbb{C}^n$ approximates an eigenvector of A . How can we use q to estimate the associated eigenvalue? Our approach will be to minimize a residual. If q is

exactly an eigenvector, then there exists exactly one number ρ for which

$$Aq = \rho q. \quad (5.3.22)$$

This number is the eigenvalue. If q is not an eigenvector, then there is no value of ρ for which (5.3.22) is satisfied. Equation (5.3.22) is in fact an overdetermined system of n equations in the single unknown ρ . Letting r denote the residual, $r = Aq - \rho q$, we can find the value of ρ for which the 2-norm $\|r\|_2$ assumes a minimum. In the case when q is an eigenvector, the minimizing ρ is exactly the associated eigenvalue. It therefore seems reasonable that if q merely approximates an eigenvector, then the minimizing ρ should at least yield a good estimate of the associated eigenvalue.

The choice of the 2-norm means that the minimization problem is a least squares problem, so we can solve it by any of the techniques developed in Chapter 3. Thus we can use a QR decomposition or the normal equations, for example. It is simplest to use the normal equations. In Chapter 3 we restricted our attention to real matrices, but all of the developments of that chapter can be carried over to the complex setting. The only modification that must be made is that whenever a transpose, say B^T , occurs, it must be replaced by the conjugate transpose B^* . This is the matrix obtained by transposing B and then taking complex conjugates of all entries. Making this modification to Theorem 3.5.21, we find that the normal equations for the complex overdetermined system $Cz = b$ are $C^*Cz = C^*b$. Rewriting (5.3.22) as

$$q\rho = Aq, \quad (5.3.23)$$

we see that the role of C is played by q , that of z is played by ρ , and that of b is played by Aq . Thus the normal equations for (5.3.23) are

$$(q^*q)\rho = q^*Aq.$$

Actually there is only one normal equation in this case, because there is only one unknown. Its solution is

$$\rho = \frac{q^*Aq}{q^*q}.$$

This number is called the *Rayleigh quotient* of q with respect to A . We have just proved the following theorem.

Theorem 5.3.24 *Let $A \in \mathbb{C}^{n \times n}$ and $q \in \mathbb{C}^n$. The unique complex number that minimizes $\|Aq - \rho q\|_2$ is the Rayleigh quotient $\rho = q^*Aq/q^*q$.*

In particular, if q is an eigenvector of A , then the Rayleigh quotient equals the associated eigenvalue. This is a consequence of Theorem 5.3.24, but it is also perfectly obvious from the form of the Rayleigh quotient.

Two other proofs of Theorem 5.3.24 are pursued in Exercise 5.3.48.

The next theorem shows that the Rayleigh quotient of q does indeed approximate an eigenvalue if q approximates an eigenvector. In the theorem the vectors are normalized so that they have Euclidean norm 1. This simplifies both the statement

and the proof of the theorem, but it is a matter of convenience, not necessity. Notice that if $\|q\|_2 = 1$, the Rayleigh quotient has the simpler form $\rho = q^* A q$.

Theorem 5.3.25 *Let $A \in \mathbb{C}^{n \times n}$ and let v be an eigenvector of A with associated eigenvalue λ . Assume $\|v\|_2 = 1$. Let $q \in \mathbb{C}^n$ with $\|q\|_2 = 1$, and let $\rho = q^* A q$ be the Rayleigh quotient of q . Then*

$$|\lambda - \rho| \leq 2\|A\|_2\|v - q\|_2.$$

Proof. Since $Av = \lambda v$ and $\|v\|_2 = 1$, $\lambda = v^* Av$. Thus $\lambda - \rho = v^* Av - q^* A q = v^* Av - v^* A q + v^* A q - q^* A q = v^* A(v - q) + (v - q)^* A q$. Therefore $|\lambda - \rho| \leq |v^* A(v - q)| + |(v - q)^* A q|$. By the Cauchy-Schwarz inequality (3.2.1), $|v^* A(v - q)| \leq \|v\|_2 \|A(v - q)\|_2 = \|A(v - q)\|_2$, and by Theorem 2.1.24,

$$\|A(v - q)\|_2 \leq \|A\|_2\|v - q\|_2.$$

Thus $|v^* A(v - q)| \leq \|A\|_2\|v - q\|_2$. Similarly $|(v - q)^* A q| \leq \|A\|_2\|v - q\|_2$. The assertion of the theorem follows. $\square$

Thus if $\|v - q\|_2 < \epsilon$, then $|\lambda - \rho| \leq 2\|A\|_2\epsilon$. This can be expressed briefly by saying that if $\|v - q\|_2 = O(\epsilon)$, then $|\lambda - \rho| = O(\epsilon)$.

Rayleigh Quotient Iteration

Rayleigh quotient iteration is that variant of inverse iteration in which the Rayleigh quotient of each q_j is computed and used as the shift for the next iteration. Thus a step of Rayleigh quotient iteration is as follows:

$$\rho_j = \frac{q_j^* A q_j}{q_j^* q_j}, \quad (A - \rho_j I)\hat{q}_{j+1} = q_j, \quad q_{j+1} = \sigma_{j+1}^{-1} \hat{q}_{j+1},$$

where σ_{j+1} is any convenient scaling factor.

Because a different shift is used at each step, it is difficult to analyze the global convergence properties of Rayleigh quotient iteration. The algorithm is not guaranteed to converge to an eigenvector, but experience suggests that it seldom fails. When it does converge, it generally converges rapidly. The next example illustrates the swift convergence of Rayleigh quotient iteration.

Example 5.3.26 Consider once again the matrix

$$A = \begin{bmatrix} 9 & 1 \\ 1 & 2 \end{bmatrix}.$$

We used Rayleigh quotient iteration starting with $q_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ to calculate an eigenvector of A . The results are listed in Table 5.3. Again we have normalized the iterates so that the first component is 1. You can easily check that q_5 is an eigenvector with associated eigenvalue ρ_5 , correct to fourteen decimal places. Notice the rapid

j	q_j (second component)	ρ_j
0	1.00000000000000	6.50000000000000
1	-0.27272727272727	8.00769230769231
2	0.22155834654310	9.09484426192450
3	0.13955130581106	9.14005316814698
4	0.14005494476317	9.14005494464026
5	0.14005494464026	9.14005494464026

Table 5.3 Rayleigh quotient iteration

convergence: q_2 has no correct digits, q_3 has (essentially) three correct digits, and q_4 has nine correct digits. $\square$

Exercise 5.3.27 Calculate the ratios $\|q_{j+1} - v\|/\|q_j - v\|$, $j = 0, 1, 2, 3$, from Table 5.3, where $v = q_5 =$ the eigenvector. Notice that the ratios decrease dramatically with increasing j . $\square$

Exercise 5.3.27 shows that the rate of convergence observed in Example 5.3.26 is better than linear. This is to be expected since the convergence rate depends on a ratio of shifted eigenvalues that improves from one step to the next.

Let us take a closer look at the convergence rate of Rayleigh quotient iteration. (However, what follows should not be mistaken for a rigorous analysis.) Let (q_j) be a sequence of vectors generated by Rayleigh quotient iteration. We shall simplify the analysis by assuming that $\|q_j\|_2 = 1$ for all j . Clearly we may choose to normalize so that this is the case. Suppose $q_j \rightarrow v_i$ as $j \rightarrow \infty$. Then also $\|v_i\|_2 = 1$. Assume further that the associated eigenvalue λ_i is not a multiple eigenvalue, and let λ_k denote the closest eigenvalue to λ_i with $k \neq i$. Since the j th step of Rayleigh quotient iteration is just a power iteration with matrix $(A - \rho_j I)^{-1}$, we know that

$$\|v_i - q_{j+1}\|_2 \approx r_j \|v_i - q_j\|_2, \quad (5.3.28)$$

where r_j is the ratio of the two eigenvalues of $(A - \rho_j I)^{-1}$ of largest absolute value. By Theorem 5.3.25 the Rayleigh quotients ρ_j converge to λ_i . Once ρ_j is close enough to λ_i , the two largest eigenvalues of $(A - \rho_j I)^{-1}$ will be $(\lambda_i - \rho_j)^{-1}$ and $(\lambda_k - \rho_j)^{-1}$. Thus

$$r_j = |(\lambda_k - \rho_j)^{-1}/(\lambda_i - \rho_j)^{-1}| = |(\lambda_i - \rho_j)/(\lambda_k - \rho_j)|.$$

By Theorem 5.3.25, $|\lambda_i - \rho_j| \leq 2\|A\|_2 \|v_i - q_j\|_2$. Also, since $\rho_j \approx \lambda_i$, we can make the approximation $|\lambda_k - \rho_j| \approx |\lambda_k - \lambda_i|$. Thus

$$r_j \approx \frac{2\|A\|_2}{|\lambda_k - \lambda_i|} \|v_i - q_j\|_2 = C \|v_i - q_j\|_2,$$

where $C = 2\|A\|_2/|\lambda_k - \lambda_i|$. Substituting this estimate of r_j into (5.3.28), we obtain the estimate

$$\|v_i - q_{j+1}\|_2 \approx C\|v_i - q_j\|_2^2. \quad (5.3.29)$$

Thus the error after $j + 1$ iterations is proportional to the square of the error after j iterations. Another way to express this is to say that if $\|v_i - q_j\|_2 = O(\epsilon)$, then $\|v_i - q_{j+1}\|_2 = O(\epsilon^2)$. If ϵ is small, then ϵ^2 is tiny. A sequence whose convergence rate satisfies (5.3.29) is said to converge *quadratically*. More specifically, we say that $q_j \rightarrow v$ *quadratically* as $j \rightarrow \infty$ if there exists a nonzero constant C such that

$$\lim_{j \rightarrow \infty} \frac{\|v - q_{j+1}\|_2}{\|v - q_j\|_2^2} = C. \quad (5.3.30)$$

The estimates that we have made above indicate that Rayleigh quotient iteration typically converges quadratically when it does converge.

A rule of thumb that one sometimes hears is this: quadratic convergence means that the number of correct digits doubles with each iteration. This is true if $C \approx 1$ in (5.3.29) and (5.3.30), for if q_j agrees with v_i to s_j decimal places, then $\|v_i - q_j\|_2 \approx 10^{-s_j}$. Thus, by (5.3.29) with $C \approx 1$, $\|v_i - q_{j+1}\|_2 \approx 10^{-2s_j}$; that is, q_{j+1} agrees with v_i to about $2s_j$ decimal places. Even if $C \not\approx 1$, the rule of thumb is valid in the limit. If $C \approx 10^t$, then $\|v_i - q_{j+1}\|_2 \approx 10^{t-2s_j}$. Thus q_{j+1} agrees with v_i to about $2s_j - t$ decimal places. As s_j grows, t becomes increasingly insignificant. Once s_j is large enough, t can be ignored.

In Example 5.3.26 the rate of convergence appears to be better than quadratic, since the number of correct digits roughly triples with each iteration. This is not an accident, rather it is a consequence of the special form of the matrix. Note that in this example A is a real symmetric matrix. For matrices of this type, the Rayleigh quotient approximates the eigenvalue better than Theorem 5.3.25 would indicate. In Section 5.4 we will see (Exercise 5.4.33) that for symmetric matrices, if $\|v - q\|_2 = O(\epsilon)$, then $|\lambda - \rho| = O(\epsilon^2)$. (The notation is that of Theorem 5.3.25.) Using this estimate instead of Theorem 5.3.25 to estimate r_j , we find that

$$\|v_i - q_{j+1}\|_2 \approx C\|v_i - q_j\|_2^3.$$

This is called *cubic* convergence. More precisely, $q_j \rightarrow v$ *cubically* as $j \rightarrow \infty$ if there is a constant $C \neq 0$ such that

$$\lim_{j \rightarrow \infty} \frac{\|v - q_{j+1}\|_2}{\|v - q_j\|_2^3} = C. \quad (5.3.31)$$

In this case the number of correct digits roughly triples with each iteration.

Exercise 5.3.32 In Example 5.3.26, the computations were performed using IEEE standard double-precision floating point arithmetic, which gives about sixteen decimal digits accuracy. If the computations had been done in quadruple precision arithmetic. About how many correct digits would you expect the iterate q_5 to have? $\square$

Quite a lot can be said about Rayleigh quotient iteration in the symmetric case that cannot be said in general. Not only is the convergence cubic when it occurs, it is also known that Rayleigh quotient iteration converges for almost all choices of starting vector. Of course the eigenvector that is reached depends on the choice of starting vector. Unfortunately there is no simple characterization of this dependence. See [54, Chapter 4] for details.

Rayleigh quotient iteration can be expensive. Since a different shift is used at each iteration, a new LU decomposition is needed at each step. This costs $O(n^3)$ flops per iteration for a full matrix and thus makes the method too expensive. However, there are some classes of matrices for which Rayleigh quotient iteration is economical. For example, a matrix A is said to have *upper Hessenberg* form if $a_{ij} = 0$ whenever $i > j + 1$. This means that the matrix has the nearly triangular form

$$\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ & * & * & * & * \\ & & * & * & * \\ & & & * & * \end{bmatrix}.$$

Exercise 5.3.33 Let A be an upper Hessenberg matrix.

- Show that an LU decomposition of A (with partial pivoting) can be accomplished in about n^2 flops. What special form does L have in this case?
- Show that a QR decomposition of A can be accomplished by applying a sequence of $n - 1$ plane rotators to A . Show that the total flop count is $O(n^2)$, assuming the factor Q is not assembled but left as a product of rotators.
- If Q were assembled, what special form would it have?

□

Exercise 5.3.33 shows that we can apply Rayleigh quotient iteration to an upper Hessenberg matrix at a cost of only $O(n^2)$ flops per iteration, which is relatively economical. Hessenberg matrices will play an important role in this chapter. In Section 5.5 we will see that the eigenvalue problem for an arbitrary matrix can be reduced to that of finding the eigenvalues (and eigenvectors, if desired) of a related upper Hessenberg matrix. We can use Rayleigh quotient iteration to attack the Hessenberg matrix at reasonable cost. However, we will not focus on Rayleigh quotient iteration. Instead we will develop a more powerful algorithm, the QR algorithm, which is based on the QR decomposition. Rayleigh quotient iterations takes place implicitly within (certain versions of) the QR algorithm.

Exercise 5.3.34 Let $A = \begin{bmatrix} 8 & 1 \\ -2 & 1 \end{bmatrix}$. Write a simple MATLAB script to perform Rayleigh quotient iteration on A .

- Use $q_0 = \begin{bmatrix} 1 & 1 \end{bmatrix}^T$ as a starting vector. Iterate until the limit of machine precision is reached. (Use `format long` to display 15 digits.) Notice that

the iterates converge to a different eigenvector than in Exercises 5.3.7 and 5.3.17. Notice also that once the iterates get close to the limit, the number of correct digits roughly doubles with each iteration. This is true of both the iteration vector and the Rayleigh quotient. Thus both converge quadratically.

- (b) Repeat part (a) with the starting vector $q_0 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$. This time the iterates converge to the same eigenvector as in Exercises 5.3.7 and 5.3.17.

□

Exercise 5.3.35 Work this problem by hand. Consider the real, symmetric matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

- (a) Apply Rayleigh quotient iteration to A with starting vector $q_0 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$. This is an exceptional case in which Rayleigh quotient iteration fails to converge.
- (b) Calculate the eigenvalues and eigenvectors by some other means and show that (i) the iterates $q_0, q_1, q_2, \dots$ exactly bisect the angle between the two linearly independent eigenvectors and (ii) the Rayleigh quotients $\rho_0, \rho_1, \rho_2, \dots$ lie exactly half way between the two eigenvalues. Thus the sequence “cannot decide” which eigenvector to approach.

□

Exercise 5.3.36 Let

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 9 & 2 \\ 0 & -1 & 2 \end{bmatrix}.$$

Use Rayleigh quotient iteration with various starting vectors to calculate three linearly independent eigenvectors and the associated eigenvalues. You can use the MATLAB script you wrote for Exercise 5.3.34.

□

Additional Exercises

Exercise 5.3.37 Each of the following sequences of positive numbers converges to zero. For each sequence (a_j) , determine whether (a_j) converges (i) slower than linearly (i.e. $\lim_{j \rightarrow \infty} a_{j+1}/a_j = 1$), (ii) linearly (i.e. $0 < \lim_{j \rightarrow \infty} a_{j+1}/a_j = r < 1$), (iii) quadratically (i.e. $\lim_{j \rightarrow \infty} a_{j+1}/a_j^2 = C \neq 0$), or (iv) cubically (i.e. $\lim_{j \rightarrow \infty} a_{j+1}/a_j^3 = C \neq 0$).

- (a) $10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}, \dots$
- (b) $10^{-1}, 10^{-2}, 10^{-4}, 10^{-8}, 10^{-16}, \dots$
- (c) $1, 1/2, 1/3, 1/4, 1/5, \dots$

(d) $10^{-3}, 10^{-6}, 10^{-9}, 10^{-12}, 10^{-15}, \dots$

(e) $0.9, 0.81, 0.729, (0.9)^4, (0.9)^5, \dots$

(f) $10^{-1}, 10^{-3}, 10^{-9}, 10^{-27}, 10^{-81}, \dots$

Notice that quadratic and cubic convergence are qualitatively better than linear convergence. However, linear convergence can be quite satisfactory if the convergence ratio r is small enough. Convergence that is slower than linear is very slow. $\square$

Exercise 5.3.38 The intent of this exercise is to show that quadratic convergence is nearly as good as cubic convergence. Suppose algorithm X produces a sequence (q_j) such that $\|q_j - v_i\| = a_j \rightarrow 0$ quadratically. Define a new algorithm Y for which one step consists of two steps of algorithm X . Then algorithm Y produces a sequence $\hat{q}_j$ for which $\|\hat{q}_j - v_i\| = b_j = a_{2j}$. Prove that $b_j \rightarrow 0$ *quartically*, that is, $\lim_{j \rightarrow \infty} b_{j+1}/b_j^4 = M \neq 0$. Thus the convergence of method Y is faster than cubic. $\square$

Exercise 5.3.39 Suppose $q_j \rightarrow v$ linearly with convergence ratio r . Show that on each iteration the number of correct decimal digits increases by about $m = -\log_{10} r$. (A rigorous argument is not required.) Thus the rate of increase in the number of correct digits is constant. $\square$

Exercise 5.3.40 The residents of Pullburg shop for groceries exactly once a week at one of three markets, named 1, 2, and 3. They are not faithful customers; they switch stores frequently. For $i, j = 1, 2, 3$, let p_{ij} be a number between 0 and 1 that represents the fraction of people who shopped at store j in a given week who will switch to store i in the following week. This can also be interpreted as the probability that a given customer will switch from store j to store i from one week to the next. These numbers can be arranged into a matrix P . Suppose the values of the p_{ij} for Pullburg are

$$P = \begin{bmatrix} 0.63 & 0.18 & 0.14 \\ 0.26 & 0.65 & 0.31 \\ 0.11 & 0.17 & 0.55 \end{bmatrix}.$$

Since $p_{23} = .31$, 31 percent of the people who shop at store 3 in a given week will switch to store 2 in the following week. Fifty-five percent of the people who shopped in store 3 will return to store 3 the following week, and so on. Notice that the sum of the entries in each column of P is one.

(a) A matrix $P \in \mathbb{R}^{n \times n}$ whose entries are nonnegative and whose column sums are all 1 is called a *stochastic matrix*. If Pullburg had n stores, the probabilities p_{ij} would form an $n \times n$ matrix. Show that regardless of what the probabilities are, the resulting $P \in \mathbb{R}^{n \times n}$ must be a stochastic matrix. Show that $w^T P = w^T$, where $w^T = [1, 1, \dots, 1]$, and 1 is an eigenvalue of P . (We call w^T a *left eigenvector* of P .) The Perron-Frobenius theory of non-negative matrices [42], [46] guarantees that, if all (or “enough”) of the p_{ij} are strictly positive, then 1 is the dominant eigenvalue of P .

- (b) Let $q_k \in \mathbb{R}^3$ be a vector whose three components represent the fraction of people in Pullburg who shopped at stores 1, 2, and 3, respectively, in week k . For example, if $q_{20} = \begin{bmatrix} 0.24 & 0.34 & 0.42 \end{bmatrix}^T$, then 24% of the population shopped in store 1, 34% in store 2, and 42% in store 3 during week 20. Show that

$$q_{k+1} = Pq_k \quad (5.3.41)$$

for all k . The sequence (q_k) is called a *Markov chain*. Equation (5.3.41) shows that we can compute a Markov chain by applying the power method with the transition matrix P and scaling factor 1.

- (c) Assuming that at week 1 one third of the population shopped at each store, calculate the fraction that visited stores 1, 2, and 3 in subsequent weeks by executing the *Markov process* (5.3.41) repeatedly. Use MATLAB to do the calculations. What is the long-term trend?
- (d) How could you have determined the long-term trend without executing the Markov process? Use MATLAB's `eig` command to determine the long-term trend without executing the Markov process.
- (e) Calculate the ratio $|\lambda_2/\lambda_1|$ to determine the rate at which the weekly state vectors q_k approach the limiting value.

□

Exercise 5.3.42 Stochastic matrices, introduced in the previous exercise, provide an example of a class of matrices for which the dominant eigenvalue tends to be significantly larger than the others.

- (a) Generate a 20×20 random stochastic matrix. For example,

```
P = rand(20);
P = P*inv(diag(sum(P,1)));
```

The first command (`rand`, not `randn`) generates a matrix with random (non-negative) entries uniformly distributed in $[0, 1]$. The second command rescales each column by dividing it by its column sum. Check that the matrix really is stochastic. Compute the eigenvalues and the ratio $|\lambda_2/\lambda_1|$ ($= \max_{j \neq 1} |\lambda_j|$).

- (b) Starting from a random starting vector $\alpha = \text{rand}(20, 1)$, compute enough iterates of the Markov chain (q_j) to observe that it converges reasonably quickly.

□

Exercise 5.3.43 This exercise is intended to convince you that if a vector q is chosen at random, then in the expansion

$$q = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$$

the coefficient c_1 is almost certain to be nonzero.

- (a) Picture the situation in $\mathbb{R}^2$. Let v_1 and v_2 be any two linearly independent vectors in $\mathbb{R}^2$. The set of all vectors $q = c_1 v_1 + c_2 v_2$ such that $c_1 = 0$ is just the subspace $\text{span}\{v_2\}$. Sketch this subspace and note that it is a very small subset of $\mathbb{R}^2$. A vector chosen at random is almost certain not to lie in it.
- (b) Repeat part (a) in $\mathbb{R}^3$.

In general any proper subspace is a very small subset of the larger space in which it lies. □

Exercise 5.3.44 Show that the sequence $(q_j) = (A^j q / \lambda_1^j)$ generated by the power method converges linearly to $c_1 v_1$ with convergence ratio $r = |\lambda_2 / \lambda_1|$, provided that $|\lambda_1| > |\lambda_2| > |\lambda_3|$, $c_1 \neq 0$, and $c_2 \neq 0$.⁶ □

Exercise 5.3.45 What happens if the power method is applied with a starting vector $q = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$ for which $c_1 = 0 \dots$

- (a) ... assuming the arithmetic is done exactly?
- (b) ... assuming roundoff errors are made?

□

Exercise 5.3.46 Suppose the power method is applied to $A \in \mathbb{C}^{n \times n}$ using some scaling strategy that produces a sequence (q_j) that converges to a dominant eigenvector. Prove that the sequence of scale factors (σ_j) converges to the dominant eigenvalue. □

Exercise 5.3.47 Let $A \in \mathbb{C}^{n \times n}$ be a matrix with eigenvalues λ and μ such that $\lambda \neq \mu$. Suppose that ρ is not an eigenvalue of A .

- (a) Show that $\|A - \rho I\| \geq |\lambda - \rho|$, where $\|\cdot\|$ denotes any induced matrix norm.
- (b) Show that $\kappa(A - \rho I) \geq |\lambda - \rho| / |\mu - \rho|$, where κ denotes the condition number with respect to any induced matrix norm.
- (c) Let (ρ_j) be a sequence of shifts such that $\rho_j \rightarrow \mu$. Prove that $\kappa(A - \rho_j I) \rightarrow \infty$ as $j \rightarrow \infty$.

□

Exercise 5.3.48 Prove Theorem 5.3.24 two different ways.

- (a) Use a complex QR decomposition of $q \in \mathbb{C}^{n \times 1}$ to prove that the Rayleigh quotient minimizes $\|Aq - \rho q\|_2$. (The factor Q in the QR decomposition satisfies $Q^* Q = I$.)

⁶It is possible to construct examples (satisfying $|\lambda_2| = |\lambda_3|$) that violate (5.3.4). These examples do, however, satisfy (5.3.3). See Exercise 5.3.11.

(b) Letting $\rho = \alpha + \beta i$, the function

$$f(\alpha, \beta) = \|Aq - \rho q\|_2^2 = \langle Aq - \rho q, Aq - \rho q \rangle$$

is a smooth function in the two real variables α and β . Use differential calculus to show that f is minimized if and only if $\rho = \alpha + \beta i$ is the Rayleigh quotient.

□

Exercise 5.3.49 Let $A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$, where $a, b \in \mathbb{R}$ and $b \neq 0$. Perform Rayleigh quotient iteration with starting vector $q_0 = \begin{bmatrix} c & d \end{bmatrix}^T$, where $c, d \in \mathbb{R}$, and $c^2 + d^2 = 1$. Analyze the problem. □

5.4 SIMILARITY TRANSFORMS

Two matrices $A, B \in \mathbb{C}^{n \times n}$ are said to be *similar* if there exists a nonsingular $S \in \mathbb{C}^{n \times n}$ such that

$$B = S^{-1}AS. \quad (5.4.1)$$

Equation (5.4.1) is called a *similarity transformation*, and S is called the *transforming matrix*. Obviously equation (5.4.1) is equivalent to

$$AS = SB. \quad (5.4.2)$$

This form is often easier to use.

As we shall see, similar matrices have the same eigenvalues, and their eigenvectors are related in a simple way. Some of the most important eigenvalue algorithms employ a sequence of similarity transformations to reduce a matrix to a simpler form. That is, they replace the original matrix by a similar one whose eigenvalues are either obvious or easily determined. This section prepares us for those algorithms by setting out some of the basic facts about similarity transformations. In the process we will cover some important material on special classes of matrices such as symmetric and orthogonal matrices and their complex counterparts.

Theorem 5.4.3 *Similar matrices have the same eigenvalues.*

Proof. Suppose A and B are similar. Then there is a nonsingular S such that $B = S^{-1}AS$. To show that A and B have the same eigenvalues, it suffices to show that they have the same characteristic polynomial. Now $\lambda I - B = S^{-1}\lambda I S - S^{-1}AS = S^{-1}(\lambda I - A)S$, so $\det(\lambda I - B) = \det(S^{-1}) \det(\lambda I - A) \det(S) = \det(\lambda I - A)$. To get this last equality we are using the facts that multiplication of complex numbers is commutative and $\det(S^{-1}) \det(S) = 1$. Thus A and B have the same characteristic polynomial. □

This proof shows that the similarity transformation preserves the *algebraic multiplicity* of the eigenvalue. That is, if μ is a root of order k of the equation

$\det(\lambda I - A) = 0$, then it is also a root of $\det(\lambda I - B) = 0$ of order k . The next theorem shows how the eigenvector is transformed.

Theorem 5.4.4 *Suppose $B = S^{-1}AS$. Then v is an eigenvector of A with eigenvalue λ if and only if $S^{-1}v$ is an eigenvector of B with associated eigenvalue λ .*

Proof. Suppose $Av = \lambda v$. Then $B(S^{-1}v) = S^{-1}ASS^{-1}v = S^{-1}Av = S^{-1}\lambda v = \lambda(S^{-1}v)$. Thus $B(S^{-1}v) = \lambda(S^{-1}v)$. Conversely, if $B(S^{-1}v) = \lambda(S^{-1}v)$, one easily deduces that $Av = \lambda v$ by a similar argument. $\square$

Exercise 5.4.5 Let D be any diagonal matrix. Find a set of n linearly independent eigenvectors of D , thereby demonstrating that D is semisimple. $\square$

The next theorem demonstrates that a matrix is semisimple if and only if it is similar to a diagonal matrix.

Theorem 5.4.6 *Let $A \in \mathbb{C}^{n \times n}$ be a semisimple matrix with linearly independent eigenvectors $v_1, v_2, \dots, v_n$ and eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Define a diagonal matrix D and a nonsingular matrix V by $D = \text{diag}\{\lambda_1, \dots, \lambda_n\}$ and $V = [v_1 \ \cdots \ v_n]$. Then $V^{-1}AV = D$. Conversely, suppose A satisfies $V^{-1}AV = D$, where D is diagonal and V is nonsingular. Then the columns of V are n linearly independent eigenvectors of A , and the main-diagonal entries of D are the associated eigenvalues. In particular, A is semisimple.*

Exercise 5.4.7 Prove Theorem 5.4.6. Work with the simpler equation $AV = VB$ instead of $B = V^{-1}AV$.

- Suppose A is semisimple, and D and V are as described. Show that the equations $Av_i = v_i\lambda_i$, $i = 1, \dots, n$, imply $AV = VD$.
- Conversely, show that the equation $AV = VD$ implies $Av_i = v_i\lambda_i$, $i = 1, \dots, n$.

$\square$

Theorem 5.4.6 is about semisimple matrices. With considerable additional effort, we could obtain an extension of Theorem 5.4.6 that is valid for all matrices. See the discussion of the *Jordan canonical form* in [29], [42], or [46].

Theorem 5.4.6 tells us that we can solve the eigenvalue problem completely if we can find a similarity that transforms A to diagonal form. Unfortunately the proof is not constructive; it does not show us a way to construct D and V without knowing the eigenvalues and eigenvectors in advance. The theorem does, however, suggest a simple form toward which we might work. For example, we might try to construct a sequence of similar matrices $A = A_0, A_1, A_2, \dots$, that converges to diagonal form.

It turns out that we will be able to carry out such a program for certain classes of matrices, but in general it is too ambitious. For one thing, it cannot succeed if the matrix is defective. Even if the matrix is semisimple, the eigenvector matrix V can be ill conditioned, which is almost as bad. Similarity transformations by ill-conditioned matrices can cause disastrous growth of errors.

Geometric View of Semisimple Matrices

Recall from elementary linear algebra that every matrix $A \in \mathbb{C}^{n \times n}$ can be viewed as a linear transformation whose action is to map $x \in \mathbb{C}^n$ to $Ax \in \mathbb{C}^n$. If A is semisimple, then its action on $\mathbb{C}^n$ is easily pictured. A has a set of n linearly independent eigenvectors $v_1, \dots, v_n$, which form a basis for $\mathbb{C}^n$. Every $x \in \mathbb{C}^n$ can be expressed as a linear combination $x = c_1 v_1 + \dots + c_n v_n$. The action of A on each $c_i v_i$ is simply to multiply it by the scalar λ_i , and the action of A on x is just a sum of such simple actions: $Ax = \lambda_1 c_1 v_1 + \lambda_2 c_2 v_2 + \dots + \lambda_n c_n v_n$.

Theorem 5.4.6 is actually just a restatement of this fact in matrix form. We will attempt to clarify this without going into detail. Recall again from elementary linear algebra that a linear transformation $A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ can be represented in numerous ways. Given any basis of $\mathbb{C}^n$, there is a unique matrix that represents A with respect to that basis. Two matrices represent the same linear transformation (with respect to different bases) if and only if they are similar. Thus a similarity transformation amounts just to a change of basis, that is, a change of coordinate system. Theorem 5.4.6 says that if A is semisimple, then there exists a coordinate system in which it is represented by a diagonal matrix. This coordinate system has eigenvectors of A as its coordinate axes.

Unitary Similarity Transformations

In Chapter 3 we introduced orthogonal matrices and noted that they have numerous desirable properties. The complex analogue of the orthogonal matrix is the unitary matrix, which was also introduced in Chapter 3. Recall that a matrix $U \in \mathbb{C}^{n \times n}$ is *unitary* if $U^* U = I$, that is, $U^* = U^{-1}$. The class of unitary matrices contains the (real) orthogonal matrices. In Exercises 3.2.51 through 3.2.59 you showed that: (1) The product of unitary matrices is unitary. (2) The inverse of a unitary matrix is unitary. (3) The complex inner product $\langle x, y \rangle = y^* x$ and Euclidean norm are preserved by unitary matrices; that is, $\langle Ux, Uy \rangle = \langle x, y \rangle$ and $\|Ux\|_2 = \|x\|_2$ for all $x, y \in \mathbb{C}^n$. (4) Rotators and reflectors have complex analogues. (5) Every $A \in \mathbb{C}^{n \times n}$ can be expressed as a product $A = QR$, where Q is unitary and R is upper triangular. You also showed (Exercise 3.4.37) that a matrix $U \in \mathbb{C}^{n \times n}$ is unitary if and only if its columns are orthonormal. Of course the orthonormality is with respect to the complex inner product.

Two matrices $A, B \in \mathbb{C}^{n \times n}$ are *unitarily similar* if there is a unitary matrix $U \in \mathbb{C}^{n \times n}$ such that $B = U^{-1}AU$. since $U^{-1} = U^*$, the similarity can also be expressed as $B = U^*AU$. If A, B , and U are all real, then U is orthogonal, and A and B are said to be *orthogonally similar*. Unitary similarity transformations have some nice properties not possessed by similarity transformations in general.

Exercise 5.4.8

- (a) Show that if U is unitary, then $\|U\|_2 = 1$ and $\kappa_2(U) = 1$.

- (b) Show that if A and B are unitarily similar, then $\|B\|_2 = \|A\|_2$ and $\kappa_2(A) = \kappa_2(B)$.
- (c) Suppose $B = U^*AU$, where U is unitary. Show that if A is perturbed slightly, then the resulting perturbation in B is of the same magnitude. Specifically, show that if $B + \delta B = U^*(A + \delta A)U$, then $\|\delta B\|_2 = \|\delta A\|_2$.

□

This exercise shows that any errors that a matrix may contain will not be amplified by subsequent unitary similarity transformations. The same cannot be said of arbitrary similarity transformations (Exercises 5.4.27 and 5.4.28).

The results in Exercise 5.4.8 are related to results from Section 3.2, in which we showed that algorithms built using well-conditioned transformation matrices, in particular rotators and reflectors, are normwise backward stable. Those results are valid for unitary matrices as well. Thus all algorithms that consist of a sequence of transformations by unitary matrices (and in particular the complex analogues of reflectors and plane rotators) are normwise backward stable.

In addition to their favorable error propagation properties, unitary similarity transformations also preserve certain desirable matrix structures. For example, consider the following theorem. (For other examples see Exercises 5.4.35 to 5.4.41.)

Theorem 5.4.9 *If $A = A^*$ and A is unitarily similar to B , then $B = B^*$.*

Proof. Since $B = U^*AU$ for some unitary U , we have $B^* = (U^*AU)^* = U^*A^*U^{**} = U^*AU = B$. □

Exercise 5.4.10 Show by example that the conclusion of Theorem 5.4.9 does not hold for general similarity transformations. □

Recall that a matrix that satisfies $A = A^*$ is called *Hermitian*. In words, Theorem 5.4.9 states that the Hermitian property is preserved under unitary similarity transformations. If we take all of the matrices in Theorem 5.4.9 to be real, we see that if A is real and symmetric and B is orthogonally similar to A , then B is also symmetric. In other words, the symmetry property is preserved under orthogonal similarity transformations. Eigenvalue problems for which the coefficient matrix is symmetric occur frequently in applications. Since symmetric and Hermitian matrices have special properties that make them easier to handle than general matrices, it is useful to have at a hand a class of similarity transformations that preserve these properties.

The next result, Schur's theorem, is the most important result of this section. It states that every (square) matrix is unitarily similar to a triangular matrix.

Theorem 5.4.11 (*Schur's Theorem*) *Let $A \in \mathbb{C}^{n \times n}$. Then there exists a unitary matrix $U \in \mathbb{C}^{n \times n}$ and an upper-triangular matrix $T \in \mathbb{C}^{n \times n}$ such that $T = U^*AU$.*

We can equally well write $A = UTU^*$. This is a *Schur decomposition* of A .

Proof. The proof is by induction on n . The result is trivial for $n = 1$. Now let us show that it holds for $n = k$, given that it holds for $n = k - 1$. Let $A \in \mathbb{C}^{k \times k}$. Let λ be an eigenvalue of A and v an associated eigenvector, chosen so that $\|v\|_2 = 1$. Let U_1 be any unitary matrix that has v as its first column. There are many such matrices: just take any orthonormal basis of $\mathbb{C}^k$ whose first member is v , and let U_1 be the matrix whose columns are the members of the basis. Let $W \in \mathbb{C}^{k \times (k-1)}$ denote the submatrix of U_1 consisting of columns 2 through k , so that $U_1 = \begin{bmatrix} v & W \end{bmatrix}$. Since the columns of W are orthogonal to v , $W^*v = 0$. Let $A_1 = U_1^*AU_1$. Then

$$A_1 = \begin{bmatrix} v^* \\ W^* \end{bmatrix} A \begin{bmatrix} v & W \end{bmatrix} = \begin{bmatrix} v^*Av & v^*AW \\ W^*Av & W^*AW \end{bmatrix}.$$

Since $Av = \lambda v$, it follows that $v^*Av = \lambda$ and $W^*Av = \lambda W^*v = 0$. Let $\hat{A} = W^*AW$. Then A_1 has the form

$$A_1 = \left[\begin{array}{c|ccc} \lambda & * & \cdots & * \\ \hline 0 & & & \\ \vdots & & \hat{A} & \\ 0 & & & \end{array} \right].$$

$\hat{A} \in \mathbb{C}^{(k-1) \times (k-1)}$, so by the induction hypothesis there exists a unitary matrix $\hat{U}_2$ and an upper-triangular matrix $\hat{T}$ such that $\hat{T} = \hat{U}_2^* \hat{A} \hat{U}_2$. Define $U_2 \in \mathbb{C}^{k \times k}$ by

$$U_2 = \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \hat{U}_2 & \\ 0 & & & \end{array} \right].$$

Then U_2 is unitary, and

$$U_2^* A_1 U_2 = \left[\begin{array}{c|ccc} \lambda & * & \cdots & * \\ \hline 0 & & & \\ \vdots & & \hat{U}_2^* \hat{A} \hat{U}_2 & \\ 0 & & & \end{array} \right] = \left[\begin{array}{c|ccc} \lambda & * & \cdots & * \\ \hline 0 & & & \\ \vdots & & \hat{T} & \\ 0 & & & \end{array} \right],$$

which is upper triangular. Let us call this matrix T , and let $U = U_1 U_2$. Then $T = U_2^* A_1 U_2 = U_2^* U_1^* A U_1 U_2 = U^* A U$. $\square$

The main-diagonal entries of T are the eigenvalues of A . Thus, if we can find the unitary similarity transformation U that transforms A to upper triangular form, then we have the eigenvalues of A . Unfortunately the proof of Schur's theorem is non-constructive; it does not give us a recipe for computing U without knowing eigenvectors in advance. Nevertheless it gives us reason to hope that we will be able to create an algorithm that produces a sequence of unitarily similar matrices $A = A_0, A_1, A_2, \dots$ that converges to upper-triangular form. Indeed there is an algorithm

that does essentially this, namely, the QR algorithm, which we will introduce in Section 5.6.

It follows easily from the equation $T = U^*AU$ that the first column of U is necessarily an eigenvector of A (regardless of how we may choose to compute U and T). Indeed, rewrite the equation as $AU = UT$, then compare first columns. The first column of AU is Au_1 , where u_1 denotes the first column of U . This must equal the first column of UT , which is u_1t_{11} . Thus $Au_1 = u_1t_{11}$, which means that u_1 is an eigenvector of A with associated eigenvalue t_{11} . In general the other columns of U are not eigenvectors of A . The extraction of the other eigenvectors requires a bit more work. See Exercise 5.4.31 and Section 5.8.

Schur's theorem is comparable in spirit to Theorem 5.4.6, which states, in part, that every semisimple matrix is similar to a diagonal matrix. Schur's theorem is more modest in the sense that the triangular form is not as simple and elegant as the diagonal form of Theorem 5.4.6. On the other hand, Schur's theorem is valid for all matrices, not just the semisimple ones. Moreover, the unitary similarity transformations of Schur's theorem are well behaved from a numerical standpoint.

A class of matrices for which Schur's theorem and Theorem 5.4.6 overlap is that of Hermitian matrices. If A is Hermitian, the matrix $T = U^*AU$ is not only upper triangular but Hermitian as well (Theorem 5.4.9). This obviously implies that T is diagonal. This result is known as the *spectral theorem* for Hermitian matrices.

Theorem 5.4.12 (Spectral Theorem for Hermitian Matrices) *Suppose $A \in \mathbb{C}^{n \times n}$ is Hermitian. Then there exists a unitary matrix $U \in \mathbb{C}^{n \times n}$ and a diagonal matrix $D \in \mathbb{R}^{n \times n}$ such that $D = U^*AU$. The columns of U are eigenvectors and the main-diagonal entries of D are eigenvalues of A .*

We can equally well write $A = UDU^*$. This is a *spectral decomposition* of A .

That the columns of U are eigenvectors follows from the last part of Theorem 5.4.6. The diagonal matrix is real because $D = D^*$. This proves that the eigenvalues of a Hermitian matrix are real.

Corollary 5.4.13 *The eigenvalues of a Hermitian matrix are real. In particular, the eigenvalues of a real symmetric matrix are real.*

Corollary 5.4.14 *Every Hermitian matrix in $\mathbb{C}^{n \times n}$ has a set of n orthonormal eigenvectors. In particular, every Hermitian matrix is semisimple.*

There are other classes of matrices for which spectral theorems analogous to Theorem 5.4.12 hold. A matrix is called *skew Hermitian* if $A^* = -A$. We are already familiar with the *unitary* matrices, which satisfy $A^* = A^{-1}$ or, equivalently, $A^*A = AA^* = I$. A matrix is *normal* if $A^*A = AA^*$. Spectral theorems for these classes of matrices are discussed in Exercises 5.4.39 through 5.4.41.

The class of normal matrices is of interest for at least two reasons. For one thing, it contains all Hermitian, skew-Hermitian, and unitary matrices, so every property that holds for normal matrices holds for these classes of matrices as well. Furthermore, this is exactly the class of matrices for which a spectral theorem holds; the following

theorem states that a matrix is unitarily similar to a diagonal matrix if and only if it is normal.

Theorem 5.4.15 (Spectral Theorem for Normal Matrices) *Let $A \in \mathbb{C}^{n \times n}$. Then A is normal if and only if there exists a unitary matrix $U \in \mathbb{C}^{n \times n}$ and a diagonal matrix $D \in \mathbb{C}^{n \times n}$ such that $D = U^*AU$.*

This theorem is proved in Exercise 5.4.41.

Corollary 5.4.16 (a) *Let $A \in \mathbb{C}^{n \times n}$ be normal. Then A has a set of n orthonormal eigenvectors.*

(b) *Conversely, if $A \in \mathbb{C}^{n \times n}$ has a set of n orthonormal eigenvectors, then A is normal.*

Exercise 5.4.17 Verify Corollary 5.4.16. □

Corollary 5.4.18 (a) *Every normal matrix is semisimple.*

(b) *Every skew-Hermitian matrix has a set of n orthonormal eigenvectors. In particular, every skew-Hermitian matrix is semisimple.*

(c) *Every unitary matrix has a set of n orthonormal eigenvectors. In particular, every unitary matrix is semisimple.*

Real Matrices

If we want to solve the general eigenvalue problem, we have to be prepared to deal with complex numbers, even if the matrix is real. There is, however, one important class of real matrices for which we can solve the entire eigenvalue problem without going outside of the real number system, namely the symmetric matrices.

Theorem 5.4.19 (Spectral Theorem for Real Symmetric Matrices) *Suppose $A \in \mathbb{R}^{n \times n}$ is symmetric. Then there exists an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ and a diagonal matrix $D \in \mathbb{R}^{n \times n}$ such that $D = U^T A U$.*

Proof. The proof is by induction on n . It is the same as the proof of Schur's theorem, except that the matrices are real and we can exploit symmetry. Therefore we just sketch the proof.

Let λ be any eigenvalue of A . Since λ is real, it has a real eigenvector v , which may be chosen so that $\|v\|_2 = 1$, associated with it. Let U_1 be a real orthogonal matrix whose first column is v , and let $A_1 = U_1^T A U_1$. Then A_1 is real and symmetric, and (as in the proof of Schur's theorem)

$$A_1 = \left[\begin{array}{c|ccc} \lambda & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \hat{A} & \\ 0 & & & \end{array} \right].$$

Since $\hat{A} \in \mathbb{R}^{(n-1) \times (n-1)}$ is symmetric, we can assume inductively that there is an orthogonal matrix $\hat{U}_2$ and a diagonal matrix $\hat{D}$ such that $\hat{D} = \hat{U}_2^T \hat{A} \hat{U}_2$. Let

$$U_2 = \left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \hat{U}_2 & \\ 0 & & & \end{array} \right], \quad D = \left[\begin{array}{c|ccc} \lambda & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \hat{D} & \\ 0 & & & \end{array} \right],$$

and $U = U_1 U_2$. Then U is orthogonal and $D = U^T A U$. $\square$

Exercise 5.4.20 Write down a detailed proof of Theorem 5.4.19. $\square$

Corollary 5.4.21 Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Then A has a set of n real, orthonormal eigenvectors.

When working with nonsymmetric matrices, we must be prepared to deal with complex numbers. Nevertheless it usually pays to delay their introduction for as long as possible, since complex arithmetic is much slower than real arithmetic. It turns out that we can bring a matrix nearly to triangular form without using complex numbers.

A matrix $T \in \mathbb{R}^{n \times n}$ is called *quasi-triangular* if it has the block upper triangular form

$$T = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1m} \\ 0 & T_{22} & & T_{2m} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & T_{mm} \end{bmatrix},$$

where each main diagonal block is either 1×1 or 2×2 , and each 2×2 block has complex eigenvalues. Thus the 1×1 blocks carry the real eigenvalues of T , and the 2×2 blocks carry the complex conjugate pairs. It is a simple matter to compute the eigenvalues of a 2×2 matrix. One simply applies the quadratic formula (carefully) to its characteristic polynomial. The Winter-Murnaghan Theorem, also known as the Real Schur Theorem states that every matrix in $\mathbb{R}^{n \times n}$ is orthogonally similar to a quasi-triangular matrix.

Theorem 5.4.22 (Wintner-Murnaghan) Let $A \in \mathbb{R}^{n \times n}$. Then there exist an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ and a quasi-triangular matrix $T \in \mathbb{R}^{n \times n}$ such that $T = U^T A U$.

The proof is like that of Schur's theorem, except that the dimension has to be reduced by two whenever we encounter a pair of complex eigenvalues. See Exercise 5.4.47 or Exercise 6.1.11.

If A is symmetric, the Wintner-Murnaghan Theorem reduces to Theorem 5.4.19. Other classes of real matrices for which the Wintner-Murnaghan Theorem simplifies nicely are the skew-symmetric ($A^T = -A$), orthogonal ($A^T = A^{-1}$), and normal ($A^T A = A A^T$) matrices. See Exercises 5.4.48 through 5.4.51.

Additional Exercises

Exercise 5.4.23 MATLAB's command $[V, D] = \text{eig}(A)$ produces a diagonal D whose main-diagonal entries are the eigenvalues of A . The associated eigenvectors are the columns of V . The matrices satisfy $AV = VD$. Normally V is nonsingular, and we have $D = V^{-1}AV$, as in Theorem 5.4.6. However, this is impossible if A is not semisimple. Try out MATLAB's eig command on the defective matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$$

Comment on the output. □

Exercise 5.4.24 The *geometric multiplicity* of an eigenvalue is defined to be the dimension of the associated eigenspace $\{v \in \mathbb{C}^n \mid Av = \lambda v\}$. Show that a similarity transformation preserves the geometric multiplicity of each eigenvalue. □

Exercise 5.4.25 Let A be a semisimple matrix. How are the geometric and algebraic multiplicities related? Recall that the algebraic multiplicity of an eigenvalue λ is equal to the multiplicity of λ as a root of the characteristic equation. □

Exercise 5.4.26 Suppose A is similar to B . Show that if A is nonsingular, then B is also nonsingular, and A^{-1} is similar to B^{-1} . □

Exercise 5.4.27 Let $S = \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix}$, where α is a real parameter.

(a) Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = S^{-1}AS$. Calculate B and conclude that $\|B\|_\infty$ can be made arbitrarily large by taking α large. (The same is true of $\|B\|_2$. We work with the ∞ -norm for convenience.)

(b) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, and $\delta A = \frac{\epsilon}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Notice that $\|\delta A\|_\infty / \|A\|_\infty = \epsilon$. Let $B = S^{-1}AS$ and $B + \delta B = S^{-1}(A + \delta A)S$. Calculate B and δB , and show that $\|\delta B\|_\infty / \|B\|_\infty$ can be made arbitrarily large by taking α large. Notice that this example is even more extreme than that of part (a), since $\|B\|_\infty$ is (asymptotically) proportional to α^2 , not α . □

Exercise 5.4.28 Suppose $B = S^{-1}AS$ and $B + \delta B = S^{-1}(A + \delta A)S$. Show that

$$(a) \quad \frac{1}{\kappa(S)} \|A\| \leq \|B\| \leq \kappa(S) \|A\|,$$

$$(b) \quad \frac{1}{\kappa(S)^2} \frac{\|\delta A\|}{\|A\|} \leq \frac{\|\delta B\|}{\|B\|} \leq \kappa(S)^2 \frac{\|\delta A\|}{\|A\|},$$

where $\|\cdot\|$ denotes any matrix norm, and κ denotes the condition number associated with that norm. Notice that the results of parts (b) and (c) of Exercise 5.4.8 are special cases of these results. $\square$

Exercise 5.4.29 Let $v \in \mathbb{C}^n$ be any nonzero vector. Show how to build a (complex) reflector whose first column is a multiple of v . Such a reflector could play the role of U_1 in the proof of Schur's theorem. $\square$

Exercise 5.4.30 Show that in Schur's theorem . . .

- (a) T can be chosen so that the eigenvalues of A appear in any desired order on the main diagonal of T .
- (b) U can be chosen so that its first column equals any desired eigenvector of A for which $\|v\|_2 = 1$. (*Hint:* Show that in the proof of Schur's theorem, the first column of U_1 equals the first column of U .)

$\square$

Exercise 5.4.31 Let $T \in \mathbb{C}^{n \times n}$ be an upper-triangular matrix with distinct eigenvalues. Sketch an algorithm that calculates a set of n linearly independent eigenvectors of T by finding solutions of equations of the form $(\lambda I - T)v = 0$. About how many flops does the whole operation require? What difficulties can arise when the eigenvalues are not distinct? This exercise will be worked out in Section 5.8. $\square$

Exercise 5.4.32 Let $A \in \mathbb{C}^{n \times n}$ be a defective matrix. Use Schur's theorem to show that for every $\epsilon > 0$, there is a semisimple matrix $A_\epsilon \in \mathbb{C}^{n \times n}$ such that $\|A - A_\epsilon\|_2 < \epsilon$. Thus the set of semisimple matrices is dense in $\mathbb{C}^{n \times n}$. $\square$

Exercise 5.4.33 In Section 5.3 it was claimed that for real symmetric matrices, the Rayleigh quotient gives a particularly good approximation to the associated eigenvalue. In this exercise we verify that claim for Hermitian matrices. Let $A \in \mathbb{C}^{n \times n}$ be Hermitian, and let $q \in \mathbb{C}^n$ be a vector satisfying $\|q\|_2 = 1$ that approximates an eigenvector v of A . Assume that $\|v\|_2 = 1$ as well. By Corollary 5.4.14, A has n orthonormal eigenvectors $v_1, \dots, v_n$, with associated eigenvalues $\lambda_1, \dots, \lambda_n$. We can choose $v_1, \dots, v_n$ in such a way that $v_1 = v$. The approximate eigenvector q can be expressed as a linear combination of $v_1, \dots, v_n$: $q = c_1 v_1 + \dots + c_n v_n$.

(a) Show that $\sum_{k=1}^n |c_k|^2 = \|q\|_2^2 = 1$.

(b) Show that $\sum_{k=2}^n |c_k|^2 \leq \|v_1 - q\|_2^2$.

(c) Derive an expression for the Rayleigh quotient $\rho = q^* A q$ in terms of the coefficients c_i and the eigenvalues λ_i .

(d) Show that $|\lambda_1 - \rho| \leq C \|v_1 - q\|_2^2$, where $C = \max_{2 \leq k \leq n} |\lambda_1 - \lambda_k|$.

(*Hint:* $\lambda_1 = \sum_{k=1}^n \lambda_1 |c_k|^2$.)

Thus if $\|v_1 - q\|_2 = O(\epsilon)$, then $|\lambda_1 - \rho| = O(\epsilon^2)$. $\square$

Exercise 5.4.34 Use Rayleigh quotients to obtain a second proof that the eigenvalues of a Hermitian matrix are real (cf. Exercise 1.4.63). $\square$

Exercise 5.4.35 Recall that a Hermitian matrix is called *positive definite* if for all nonzero $x \in \mathbb{C}^n$, $x^*Ax > 0$. Prove that if A is positive definite and B is unitarily similar to A , then B is also positive definite. $\square$

Exercise 5.4.36 Use Rayleigh quotients to prove that the eigenvalues of a positive definite matrix are positive. The next exercise indicates a second way to prove this. $\square$

Exercise 5.4.37 Let $A \in \mathbb{C}^{n \times n}$ be a Hermitian matrix. Use Theorem 5.4.12 and the result of Exercise 5.4.35 to prove that A is positive definite if and only if all of its eigenvalues are positive. $\square$

Exercise 5.4.38 A Hermitian matrix $A \in \mathbb{C}^{n \times n}$ is *positive semidefinite* if $x^*Ax \geq 0$ for all $x \in \mathbb{C}^n$. Formulate and prove results analogous to those of Exercises 5.4.35 to 5.4.37 for positive semidefinite matrices. $\square$

Exercise 5.4.39 A matrix $A \in \mathbb{C}^{n \times n}$ is *skew Hermitian* if $A^* = -A$.

- Prove that if A is skew Hermitian and B is unitarily similar to A , then B is also skew Hermitian.
- What special form does Schur's theorem (5.4.11) take when A is skew Hermitian?
- Prove that the eigenvalues of a skew Hermitian matrix are purely imaginary; that is, they satisfy $\bar{\lambda} = -\lambda$. Give two proofs, one based on Schur's theorem and one based on the Rayleigh quotient.

$\square$

Exercise 5.4.40

- Prove that if A is unitary and B is unitarily similar to A , then B is also unitary.
- Prove that a matrix $T \in \mathbb{C}^{n \times n}$ that is both upper triangular and unitary must be a diagonal matrix. (You will prove a more general result in Exercise 5.4.41.)
- What special form does Schur's theorem (5.4.11) take when A is unitary?
- Prove that the eigenvalues of a unitary matrix satisfy $\bar{\lambda} = \lambda^{-1}$. Equivalently; $\lambda\bar{\lambda} = 1$ or $|\lambda|^2 = 1$; that is, the eigenvalues lie on the unit circle in the complex plane. Give two proofs.

$\square$

Exercise 5.4.41 A matrix $A \in \mathbb{C}^{n \times n}$ is *normal* if $AA^* = A^*A$.

- Prove that all Hermitian, skew-Hermitian, and unitary matrices are normal.

- (b) Prove that if A is normal and B is unitarily similar to A , then B is also normal.
- (c) Prove that a matrix $T \in \mathbb{C}^{n \times n}$ that is both upper triangular and normal must be a diagonal matrix. (*Hint:* Use induction on n . Write T in the partitioned form

$$T = \begin{bmatrix} t_{11} & s^* \\ 0 & \hat{T} \end{bmatrix},$$

then write down the equation $TT^* = T^*T$ in partitioned form and deduce that $s = 0$ and $\hat{T}$ is also (triangular and) normal.)

- (d) Prove that every diagonal matrix is normal.
- (e) Prove Theorem 5.4.15: A is normal if and only if A is unitarily similar to a diagonal matrix.

□

Exercise 5.4.42 Let $D \in \mathbb{C}^{n \times n}$ be a diagonal matrix. Show that . . .

- (a) D is Hermitian if and only if its eigenvalues are real.
- (b) D is positive semidefinite if and only if its eigenvalues are nonnegative.
- (c) D is positive definite if and only if its eigenvalues are positive.
- (d) D is skew Hermitian if and only if its eigenvalues lie on the imaginary axis of the complex plane.
- (e) D is unitary if and only if its eigenvalues lie on the unit circle of the complex plane.

□

Exercise 5.4.43 Let $A \in \mathbb{C}^{n \times n}$ be a normal matrix. Show that . . .

- (a) A is Hermitian if and only if its eigenvalues lie on the real axis.
- (b) A is positive semidefinite if and only if its eigenvalues are nonnegative.
- (c) A is positive definite if and only if its eigenvalues are positive.
- (d) A is skew Hermitian if and only if its eigenvalues lie on the imaginary axis.
- (e) A is unitary if and only if its eigenvalues lie on the unit circle.

□

Exercise 5.4.44 Verify that the results of Exercise 5.4.33 can be carried over verbatim to normal matrices. Thus the good approximation properties of the Rayleigh quotient hold for normal matrices in general, not just Hermitian matrices.

□